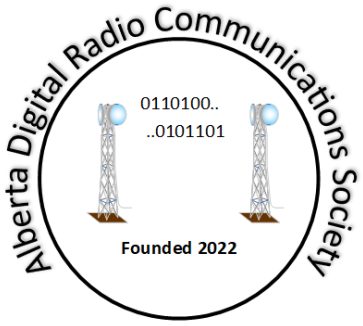


Digital Radio Projects

IP400 Unified Firmware Platform
for OFDM based nodes

Document Number: IP400-UFP-OFDM
Revision: 1.1
Status: Released
Written By: Martin C. Alcock, M. Sc, MIEEE, VE6VH

Table of Contents

Revision Status vii
Reference Documents vii
Hardware Release Status vii
Firmware Release Status vii
Intellectual Property Notice viii
Disclaimer viii
Introduction 1
  Roadmap 2
  Data Modes 3
Hardware Platform 4
Firmware Description 7
  General Structure 7
  Features 8
  KISS Serial Interface 9
  IP400 Frame Processor 14
  Platform Support 20
  Modems and Physical I/O 22
OFDM Transmissions 25
  Superframe format 25
  Data Frame 25
  IP400 Frame 26
FPGA Digital Audio Ports 27
  STM32F722 Interfaces 28
  Pi SPI Interfaces 34
Operating the Software 40
  Main Menu 40
  Setup Parameters 41
  Mesh Table 42
  Chat/Echo Mode 43
  Diagnostics Menu 45
  Send Beacon 49
  Radio Setup Menus 51
  Station Setup Menu 52
Building the Code 53
  Downloading 53
Support Software 54
  Remote Ethernet System 54
  FPGA Flash Programmer 55
  UDP to SPI software 56
  UDP Packet Generator 57

List of Tables

Table 1 Revision status vii
Table 2 Reference Documents vii
Table 3 Release Status vii
Table 4 Firmware Release Status vii
Table 5 Data Modes 3
Table 6 OFDM data rates 3
Table 7 Supported platforms 4
Table 8 Expansion Connector Pinout 6
Table 9 FPGA expansion connector 6
Table 10 Firmware Features and Module 8
Table 11 Kiss framing characters 9
Table 12 KISS frame format 9
Table 13 Kiss Commands. 9
Table 14 Test modes 11
Table 15 Flash update command arguments 11
Table 16 AX25 Frame Format 12
Table 17 AX.25 Address field format 12
Table 18 Address fields coding 12
Table 19 Command/Response bit coding 13
Table 20 Local Frame Processing 14
Table 21 Mesh Table Contents 15
Table 22 Header Symbol Coding 15
Table 23 VPN address coding 16
Table 24 Callsign character coding 16
Table 25 Frame Header Flag field definitions 17
Table 26 Flag field values 17
Table 27 Hop table contents 18
Table 28 Hop Table Flag field definitions 18
Table 29 Symbol subcarrier mapping 18
Table 30 Radio Parameters 20
Table 31 Data mod values for OFDM-AB standards 20
Table 32 Data Symbols size definitions 20
Table 33 TLV320AIC3104 Code specifications 21
Table 34 Clock generator outputs 21
Table 35 LED Usage by platform 23
Table 36 OFDM-AB Modem timing 24
Table 37. Sync code fields 26
Table 38 Randomizer Polynomials 26
Table 39 OFDM-AB SPI Interface 28
Table 40 SPI Control word 29
Table 41 CPU SPI Bank addresses 29
Table 42 OFDM-AB SPI Control and Status registers 30

Table 43 Read Register Values 30
Table 44 Write Register values 30
Table 45 Expansion Registers 30
Table 46 Register Abbreviations 31
Table 47 Carrier detection configuration 31
Table 48 FIFO change of state interrupts 32
Table 49 STM32F722 FPGA GPIO usage 32
Table 50 CPU SPI Bank addresses 34
Table 51 SPI Frame Header format 35
Table 52 Frame flags definition 35
Table 53 Fragmentation field values 36
Table 54 Packet coding 36
Table 55 Modem qualifier fields 36
Table 56 SPI to IP400 Header Mapping 37
Table 57 CTCSS encode/decoder register definitions 38
Table 58 Control/Status register definitions 38
Table 59 Register Bit definitions 38
Table 60 CTCSS Tone encoder divider values 39
Table 61 Goertzel Detection algorithm coefficients 39
Table 62 Main Menu Items 40
Table 63 Mesh Table explanation 42
Table 64 Capabilities field 42
Table 65 Chat mode contol keys 43
Table 66 Diagnostics Menu 45
Table 67 Frame Statistics 46
Table 68 Frame Processor Statistics 47
Table 69 SPI Statistics 47
Table 70 Codec test menu 48
Table 71 Beacon Frame Payload 49
Table 72 LED Test Cycling 49
Table 73 Radio Setup Meny Item 51
Table 74 Station setup parameters 52
Table 75 Code Components 53
Table 76 FPGA programmer command line options 55
Table 77 Pi utility software command parameters 56
Table 78 Pi Utility debug levels 56
Table 79 Loopback mode arguments 56
Table 80 Command line parameters for UDP packet generator 57
Table 81 Generated frame types 57

List of Figures

Figure 1 IP400 Node Development Roadmap 2
Figure 2 Advanced Mesh Network Controller 4
Figure 3 DB15 Connector Pinout 5
Figure 4 IP400 System General Structure 7
Figure 5 Use of transposition in KISS frames 10
Figure 6 IP400 Frame Processor 14
Figure 7 Frame Fragmentation Schemes 19
Figure 8 OFDM Transceiver Stack 22
Figure 9 Superframe format 25
Figure 10 Data Frame Format 25
Figure 11 IP400 Frame format 26
Figure 12 FPGA Digital Audio Ports 27
Figure 13 SPI Interface timing 28
Figure 14 Transmitter state machine 33
Figure 15 Pi SPI Interfaces 34
Figure 16 Main Menu 40
Figure 17 Setup Parameters 41
Figure 18 Mesh Table 42
Figure 19 Diagnostics Menu 45
Figure 20 OFDM-AB Modem Statistics 46
Figure 21 Codec Tests Menu 48
Figure 22 GPS NMEA sentences 50
Figure 23 OFDM-AB Radio Setup Menu 51
Figure 24 List of Radio Settings 51
Figure 25 Station Setup Menu 52
Figure 26 Ethernet Subsystem 54
Figure 27 Flash programming protocol 55

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Revision Status

Revision	Date	Description
1.0	Sept, 2026	First Release
1.0a	Oct, 2026	Minor update to KISS commands
1.1	Oct, 2026	Added Superframe details

Table 1 Revision status

Reference Documents

Author	Issue Date	Document Number	Description
M. Alcock	Aug 2026	IP400-UFP-FSK	IP400 UFP for Legacy Nodes
M. Alcock **R. Carrillo**	Feb 2026	IP400-OFDM	IP400 OFDM-AB Data Mode

Table 2 Reference Documents

Hardware Release Status

Platform	Processor	Status	Base Version	Release Date
Nucleo CC2	STM32WL33	No longer manufactured	2.0	Q3, 2026
Mini Node	STM32WL33			
AMNC	STM32F722 FPGA	In development	2.1	Q4, 2026

Table 3 Release Status

Firmware Release Status

Release Date	Version	Status	Features
Q3, 2026	2.0	Released	UFP for Mini node with KISS, SPI, 4FSK mode
Q4, 2026	2.1	Beta	Add AMNC with KISS, SPI, OFDM-AB up to QAM-64, all BW, CRC only
Q1, 2027	2.1a	Planned	FPGA upgrade and addition of ethernet subsystems
Q2, 2027	2.2	Planned	Add QAM-128, QAM-256 and FEC using Trellis coding

Table 4 Firmware Release Status

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Introduction

The IP400 project was launched to experiment with digital mesh networking on the 400 MHz band, using commercial off the shelf devices. To get the project rolling, the initial firmware was developed for an STM32WL33 microcontroller [2], on several platforms that support the basic mode.

As the project has matured, new data modes have been explored, to satisfy the stated project goals. At the ZRDC conference in September of 2025, two of the stated goals were to implement a mode that could be retrofitted to existing repeater systems and off the shelf radios, complete with a mobile TNC terminal. Several modes were investigated, including a derivative of the Digital Mobile Radio (DMR) protocol [3], which led to developing the concept of an audio band variant of OFDM for this purpose, dubbed OFDM-AB.

OFDM (Orthogonal Frequency Division Multiplexing) is a multi-carrier system where data is spread over a series of carriers instead of a single carrier. Each carrier is modulated individually, and they can be independent of each other. The modulation methods employed use the familiar BPSK, QPSK and QAM constellations, so standard coding and decoding techniques can be employed. Some carriers are designated as ‘pilot’ carriers, which are coded with known data bits.

The challenge of introducing OFDM to amateur radio is to be able to retrofit existing radios with a minimum of modifications by using the existing microphone and speaker connections, and to be able to ‘undo’ any effects that are introduced by the radio and channel characteristics to accurately decode any received data.

This was accomplished by matching the occupied carrier bandwidth with that of the radio, and in addition the use of a known sequence which could be used to train an equalizer to remove radio and channel anomalies. Testing on various radio types precipitated four different modes, from a lowest common denominator that would work with any radio, up to a higher bandwidth mode that would be limited to specialized radios. Data rates vary between modes based on the constellation employed, at the lowest order of 3 bits/symbol, raw data rates from 7,260 to 19,360 bits/sec could be achieved, and at the highest order 21,384 to 57,024 can be realized, depending on the available signal to noise ratio. These are the highest data rates that have ever been achieved on ‘off the shelf’ amateur radio equipment.

This document describes the firmware that has been developed using the OFDM technology base, and its application to two node types: the first the new Advanced Mesh Network Controller which connects directly to a radio and can be used in a standalone manner or as a Pi HAT for a repeater upgrade or to be accessed remotely over ethernet.

Roadmap

Figure 1 illustrates the roadmap for the IP400 nodes and firmware development. This document describes the third phase additions to the project.

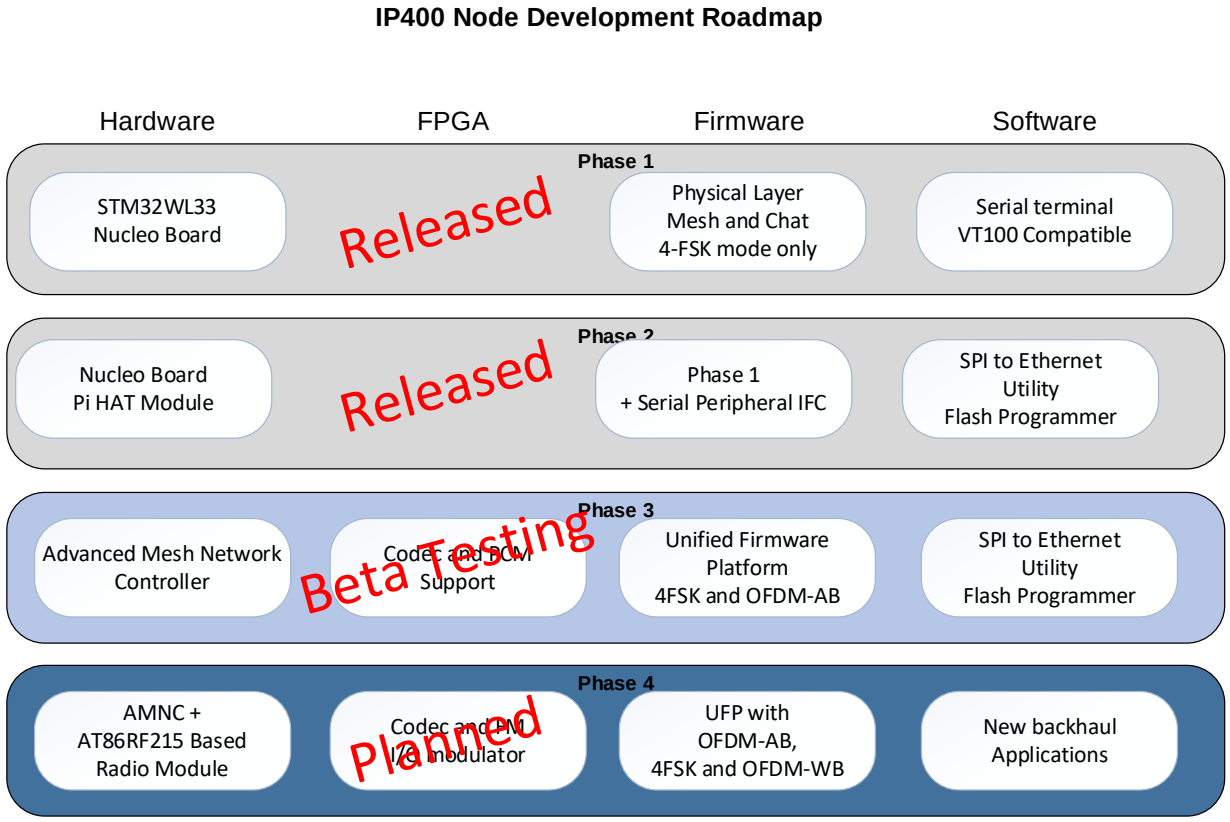


Figure 1 IP400 Node Development Roadmap

Future plans for phase 4 include adding a radio module onto the AMNC that is compatible with the mini node, and also has provisions for OFDM in the 400MHz band, with speeds up to 2.4Mb/s.

Data Modes

The four data modes are designated, narrowband (NB), enhanced narrowband (ENB), wideband (WB) and enhanced wideband (EWB). Each mode is designed for a specific radio application, as described in Table 5.

Index	Mode	Audio Bandwidth	Max Data Rate[1]	Application
0	NB	2,625	19,412	Most FM radios using mic and speaker connections.
1	ENB	3,656	26,824	The majority of FM radios using mic and speaker, except those with sharp high pass filters.
2	WB	5,766	42,353	Radio with a data port, or FM radios modified with direct discriminator varactor connections.
3	EWB	7,781	57,176	High end radios with extended audio passbands

Table 5 Data Modes

The data rates vary depending on the modulation scheme employed and signal to noise ratios. Maximum data rates should be considered for point to point links, others for general use. Table 6 illustrates the schemes used and the maximum data achievable in each mode.

Index	Method	S/N Ratio	Fraction Of Highest	NB	ENB	WB	EWB
0	BPSK	Lowest	0.13	2,426	3,353	5,294	7,147
1	QPSK		0.25	4,853	6,706	10,588	14,294
2	QAM-8		0.38	7,279	10,059	15,882	21,441
3	QAM-16		0.50	9,706	13,412	21,176	28,588
4	QAM-32		0.63	12,132	16,765	26,471	35,735
5	QAM-64		0.75	14,559	20,118	31,765	42,882
6	QAM-128		0.88	16,985	23,471	37,059	50,029
7	QAM-256	Highest	1.00	19,412	26,824	42,353	57,176

Table 6 OFDM data rates

[1] Using a QAM-256 constellation

Hardware Platform

Table 7 describes the various hardware platforms, radio modes and host type and interfaces.

Platform	Processor(s)	Host and Interfaces	Data Modes	Radios Supported
AMNC Module	STM32F722 FPGA	Raspberry Pi SPI USB serial bus	All	External (2)

Table 7 Supported platforms

The Advanced Mesh Network Controller is the primary stand-alone modem that can support 2 radios simultaneously. It can be used in a 'standalone' mode, where it is powered by a USB connection and supports the readily available AX.25 KISS protocol [4], or as a HAT module for a raspberry Pi to enable a data mode for repeater systems. In this mode is it powered from a 12v supply, the USB is disabled but the Pi SPI and UARTs are used instead. Figure 2 illustrates the controller.



Figure 2 Advanced Mesh Network Controller

Connector Pinouts

Apart from the PI connector, there are three connectors on the AMNC module. The first is a compact DB-15 connector for the two radio ports, Figure 3 Illustrates the pinout of the radio connector.

Pin	Connection	Pin	Connection
1	Audio Out Port 1	**8**	Port 2 PTT
2	Audio Out Port 2	**9**	Power Ground
3	Port 1 Carrier Detect	**10**	Port 2 Carrier Detect
4	Power Ground	**11**	Port 1 Audio In
5, 15	+12V	**12**	Port 2 Audio In
6	Radio Ground	**13**	Port 1 PTT
7	Radio Ground	**14**	Power Ground

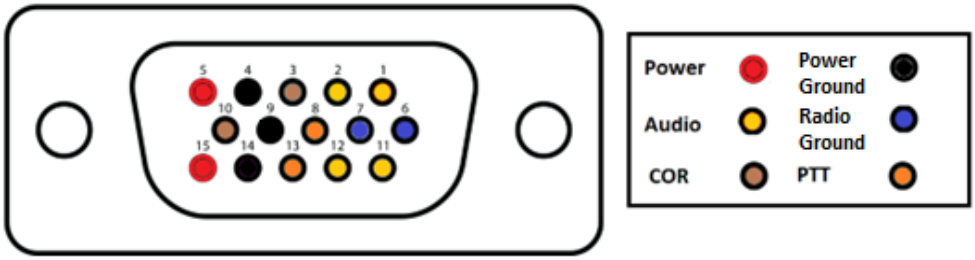


Figure 3 DB15 Connector Pinout

Note: The Power grounds (4, 9, 13) should only be used for power supply connections and only in the Pi HAT mode, to power the board and the Pi. In the USB mode, these are not connected.

All radio connections must reference the radio grounds (6, 7).

At the opposite end of the board, is a 2x12 2.0mm spaced header for debug and expansion purposes, the pinout by function is listed in Table 8.

Function	Pin	Purpose	Pin	Purpose
Power	2	+3.3v	7,8,16,24	GND
FPGA JTAG	1	TDO	3	TCK
	4	TDI	5	TMS
	6	FPGA Reset	21	FPGA_SS
CPU Debug	15	SWCLK	17	SWDIO
	13	SWUART_RX	15	SWUART_TX
	14	CPU Reset	-	
Test Points	10	TP1	12	TP2
Expansion	17	Ext Reset	18	SPI_SCK
	19	SPI_MOSI	20	SPI_MISO
	22	SPI_CSN	23	Ext CLK

Table 8 Expansion Connector Pinout

The other connector is reserved for future connections to the FPGA. Table 9 lists the pinout.

Pin	Function	Pin	Function	Pin	Function	Pin	Function
1	FPGA-52	**4**	FPGA-51	**7**	FPGA-33	**10**	FPGA-32
2	GND	**5**	FPGA-41	**8**	FPGA-40	**11**	FPGA-29
3	FPGA-44	**6**	FPGA-43	**9**	FPGA-30	**12**	+3.3V

Table 9 FPGA expansion connector

Firmware Description

General Structure

The general structure of an IP400 OFDM node is illustrated in Figure 4.

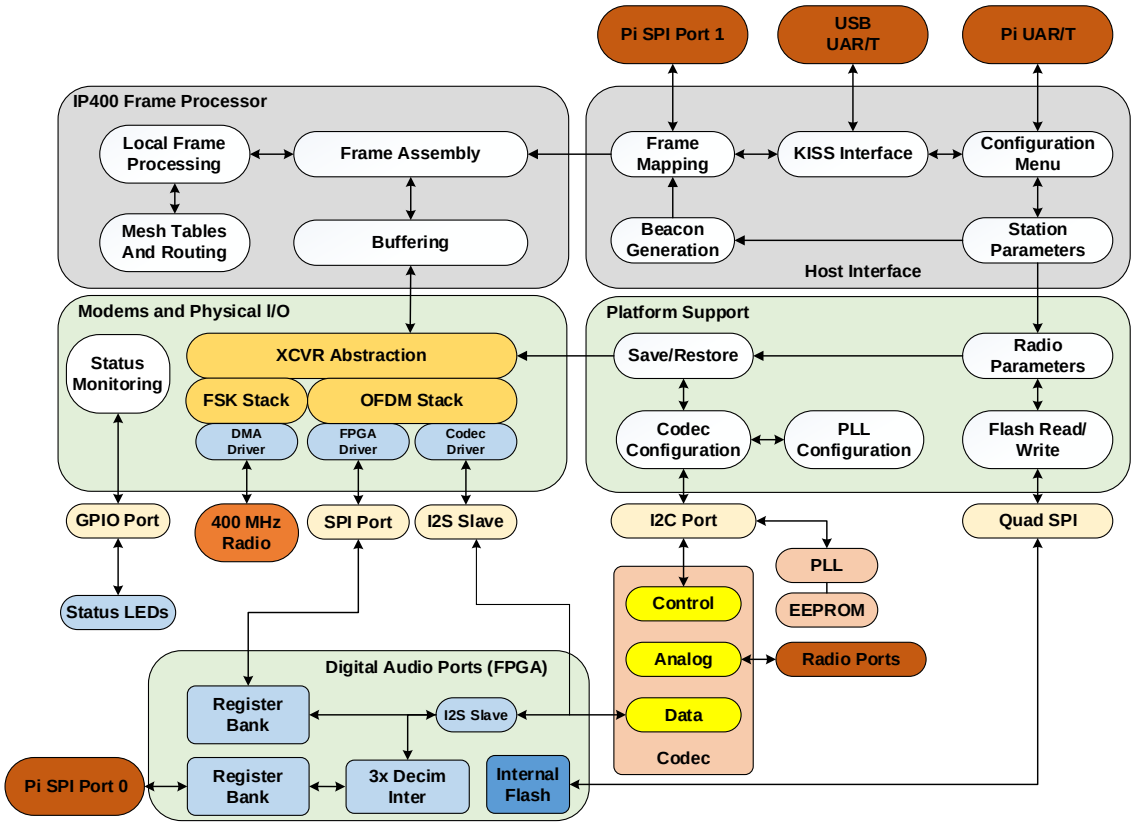


Figure 4 IP400 System General Structure

The firmware is structured in four sections:

1. IP400 Frame Processor. Assembles, sends and receives IP400 frames, processes mesh tables, interprets local frames, reassembles received frames and processes forward error correction.
2. Host Interface. Sends and receives frames an asynchronous serial interface in KISS format, or a Serial Peripheral Interface from another processor. Contains configuration interface and station parameters and generates beacon frames. Maps frames to IP400 format.
3. Modems and Physical I/O. Modem handlers for internal and external modems, and interfacing to FPGA co-processors, status monitoring and LED displays.
4. Platform Support. Setup and monitoring of devices such as Codecs and clock generators, and setup parameters for the radio modules, flash programming and permanent setup storage.

Features

Table 10 lists the firmware features by subsystem.

Sub System	Feature	Description	AMNC	RPi
Host Interface	Configuration Menu	An intuitive menu to set up parameters such as frequency, mode, power, callsign, etc.	✓	
	Beaconing	A station identification message sent periodically to all stations	✓	
	AX.25 KISS mode	A serial interface that accepts and AX.25 frames in KISS format	✓	
	Serial Peripheral Interface (SPI)	A high speed serial interface to a host processor that accepts IP frames	✓	
Ethernet	Telnet Server	Menu and Kiss interface access		✓
	TFPT Server	Firmware updates		✓
	UDP Server	IP400 or AXUDP packets		✓
	USRP Server	Digital Audio Packets		✓
IP400 Frame Processor	Mesh table	A collection of all stations heard and routing information	✓	
	Chat mode	A simple chat application	✓	
	Ping mode	A dedicated message to request a response from a remote station.	✓	
	Repeat (Digipeater)	An ability to repeat frames based on station callsigns.	✓	
Platform Support	Real Time Clock	A real time clock with a resolution of 10 seconds	✓	
	TLV320 Codec	Dual channel audio codec and radio port	✓	
	Clock Generator	Accurate PLL-based clock generator for data transmission	✓	
Modems And I/O	Status and LED display	A series of LEDs to indicate the node status	✓	
	Internal Modem	OFDM-AB	✓	
	External Modem	Interfacing for an external modem	✓	
Digital Audio Ports	Radio port	24 Kb/s 16 bit PCM for modems	✓	
	Audio port	8Ks/s 16 bit PCM audio port	✓	

Table 10 Firmware Features and Module

KISS Serial Interface

The KISS [4] serial interface connects directly to the serial port of the processor to provide an alternate data path for the audio band OFDM modem, using an asynchronous serial interface. The serial format is 1 start bit, 1 stop bit, 8 data bits, no parity, 1 bit/baud, baud rate 115200 symbols/second. KISS defines four special characters in its alphabet for framing purposes, as listed in Table 11 and the general form of a KISS frame is illustrated in Table 12. C notation is used to denote hexadecimal values. Both the Pi and USB Serial interfaces support the KISS mode but are mutually exclusive.

Character	Hex Code	Purpose
FEND	0xC0	Delimits the beginning and end of a frame
FESC	0xDB	Escape, indicates the subsequent character is an escape code
TFEND	0xDC	Transposed (escaped) frame end
TFESC	0xDD	Transposed frame escape

Table 11 Kiss framing characters

Begin	Command	Data 0…n	End
FEND	High Nibble – Port Index Low Nibble - Command	Data	FEND

Table 12 KISS frame format

KISS Commands

The supported command types are listed in Table 13.

Command Bytes	Frame Type	Argument Length	Description
00 […]	Data frame	1-1024	Sends a data frame
01	Tx Delay	1	Sets the transmit leader length
02	Persistence	1	Sets the persistence value
03	Slot Time	1	Sets the persistence wait time
04	Tx Tail	1	Sets the transmit tail timer
05	Duplex	1	Unused; reserved for future use
06 P S	Set hardware	2	
07	Set Configuration		
08	Get Version	0	Responds with firmware version number
09 T	Test mode	1	Initiates an internal test mode
0A	Set Serial number	0	Reserved for future use
0B	Get all parameters	0	Displays critical parameters
0C	Reserved		TBD
0D	Update Flash	1	Updates FPGA flash memory
0E	Get serial number	0	Returns the 16-byte serial number

Table 13 Kiss Commands.

Send Data Frame (00)

All data frames are sent in AX.25 format. In the event that a data packet contains the FEND or FESC characters, the transposed mode is used instead. Figure 5 illustrates an example where the FEND (0xC0) and FESC (0xDB) are included as data bytes. Figure 5 illustrates the format of a data frame, and the escape sequence for sending framing characters as data. The data portion cannot exceed the maximum IP400 payload size of 1024 bytes.

Send the bytes 0xC0, 0xDB out of TNC port 0

C0 - FEND	00 - DATA FRAME: port 0	DB - FESC	DC - TFEND	DB - FESC	DD - TFESC	C0 - FEND

Figure 5 Use of transposition in KISS frames

Set Transmit Delay (01)

Sets the transmitter key up delay in 10 ms units. The default start-up value is 50 (i.e., 500 ms).

Set Persistence (02)

Sets the persistence parameter, p, scaled to the range 0 - 255 with the following formula, P = p * 256 – 1. The default value is P = 63 (i.e., p = 0.25.

Set Slot Time (03)

Sets the slot interval in 20 ms units. The default is 2 (i.e., 40ms).

Set Transmit Tail (04)

Sets the time to hold up the TX after the Frame has been sent, in 20 ms units.

Set Duplex (05)

Sets the duplex mode, 0 for half duplex, nonzero for full duplex. Full duplex is not supported.

Set Hardware Command (06)

These commands are specific to a particular hardware implementation.

Set Test Mode (08)

Three different test modes are implemented as listed in Table 14.

Mode	Test
0	Turns all test modes off
1	Initiates an unmodulated CW signal at the transmitter
2	Sends a fixed tone of 1500Hz.

Table 14 Test modes

Reserved (0C)

TBD.

Flash Update (0D)

There are three possible arguments to the flash update command, as listed in Table 15.

Argument	Flash Mode
37	Reserved
E8	Initiates a boot image write for the FPGA flash ROM

Table 15 Flash update command arguments

Frame Mapping

The mapping of AX25 to IP400 frames is a process that provides a transparent and lossless translation from one format to the other. According to the Tucson Amateur Packet Radio Group [5], an AX25 data frame is made up of three fields, as illustrated in Table 16.

Address	Control	Payload
14 to 28 bytes	2-3 bytes	0-1024 bytes

Table 16 AX25 Frame Format

The address field is mapped into an IP400 MAC header, the control and payload fields are send verbatim.

Address Field Format

The address field consists of two sub-fields, the first field is 14 bytes in length for peer-to-peer communications, the second subfield contains up to two repeater addresses, which can be 0, 7, or 14 bytes in length. Table 17 illustrates the header fields.

Byte Index	Contents	Format/Field Contents	Required/Optional
0-6	Destination Callsign	ASCII value AAAAAAA0	Required Fields
7	Destination SSID	CRRSSID0	
8-13	Source Callsign	ASCII value AAAAAAA0	
14	Source SSID	CRRSSIDE	
15-20	Repeater 1 Callsign	ASCII value AAAAAAA0	Optional
21	Repeater 1 SSID	HRRSSIDE	
22-27	Repeater 2 Callsign	ASCII value AAAAAAA0	Optional
28	Repeater 2 SSID	HRRSSIDE	

Table 17 AX.25 Address field format

The fields are illustrated in Table 18.

Field Designation	Contents
AAAAAAA	7-bit subset of the ascii character code, from 00 to $7F_{16}$
C	Command/Response bit. See Table 19.
RR	Reserved bits. Set to 1
SSID	4-bit SSID address
E	Address extension bit. Set to 1 to indicate the end of the field.
H	Has been repeated. Set to one only when the frame has been repeated.

Table 18 Address fields coding

The Command/Response bit coding identifies the frame as a command or response. The bit coding is illustrated in Table 19.

Frame Type	Destination SSID C-Bit	Source SSID C-Bit
Command	1	0
Response	0	1

Table 19 Command/Response bit coding

Mapping Rules

The following are the mapping rules for AX.25 frames:

1. The callsign fields are compressed into four-bytes using excess-40 notation.
2. The SSID, H and C bits are contained in the VPN address, as shown in Table 23.
3. The MAC addresses are made up from the source and destination addresses.
4. The repeater addresses, if present, are coded into the hop table.
5. The flag fields are set as follows:
    a. Hop Count, 1 or 2 depending on source frame.
    b. Coding, constant '0110', for an encapsulated AX.25 frame.
    c. SSID – constant 0.
    d. AX.25/VPN constant '1'.
    e. Source call, dest call and repeat bits, all set to zero.

All data from the end of the address field to the end of the frame is copied verbatim to the payload portion of the IP400 frame, after the hop table, if present.

IP400 Frame Processor

Figure 6 illustrates the frame processor module from the generic structure diagram.

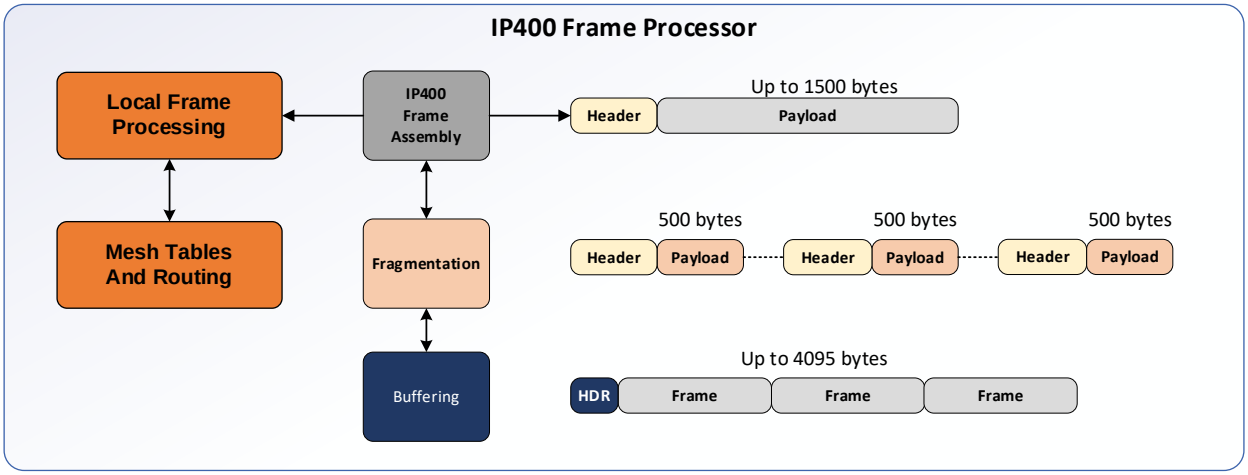


Figure 6 IP400 Frame Processor

Local Frame Processing

Table 20 illustrates the packet coding and where they are routed for processing. There are 16 packet types defined, the first 8 are processed internally, the remaining types are available for user applications, which are routed to the SPI interface by default.

AX.25 frames can only be originated at the KISS interface and are routed back to it on receive. Cross-modes between KISS and SPI are currently not supported and may produce unpredictable results.

The command frame is reserved for future use as a 'DNS lookup' mode. The intent is to be able to find the routing information to a specific node by using this frame type. Implementation is still pending.

Packet Type	Coding	Purpose	Application/Destination
UTF8_TEXT_PACKET	0	Text packet	Chat
DATA_PACKET	1	Data packet	VPN Specific; KISS or SPI
BEACON_PACKET	2	Mesh Discovery	Mesh tables
AX_25_PACKET	3	Encapsulated AX.25 packet	KISS interface
IP_ENCAPSULATED	4	Encapsulated IP packet	SPI Interface
ECHO_REQUEST	5	Request for a response	Chat
ECHO_RESPONSE	6	Response to a request	Chat
LOCAL_COMMAND	7	Routing enquiry	Mesh Tables
USER_DEFINED	8-15	Undefined	SPI Interface

Table 20 Local Frame Processing

Mesh Tables and Routing

Field	Data Type	Source
Status	MeshTableStatus	Beacon frame, TOD clock
macAddr	IP400 MAC address	Beacon frame
NextSeq	Integer	Last frame
lastRSSI	Integer	Receiver
txPower	Integer	Beacon Frame
Capabilities	SETUP_FLAGS	Station capabilities
LastHeard	TIMEOFDAY	Last time heard
hopCount	Integer	Number of hops required
RptAddr	IP400 MAC address	Repeater addresses (max 2)

Table 21 Mesh Table Contents

Frame Assembly

Each IP400 frame contains a header and potentially a hop and multipoint table which are the same size. There is no forward error correction in the header, just a CRC-16 checksum to verify validity. The fields are listed in Table 22.

Field	Size bytes)	1. Purpose in Header Symbol
Source MAC Address	6	Source MAC address
Destination MAC Address	6	Destination MAC address
Flags	2	Frame flags
Payload length LSB	1	Number of payload packets
Sequence number	4	Packet sequence number
Source address extension	4	Last 6 bytes of source callsign
Dest. address extension	4	Last 6 bytes of destination callsign
Spare	1	Not used
LRC-8	1	LRC-8 Checksum
Total Bytes	29	

Table 22 Header Symbol Coding

MAC Address Coding

The MAC addresses is composed of the first 6 bytes of the callsign compressed into 4 bytes, followed by a unique station identifier (USI). Table 23 illustrates the four methods of coding a station address. There are two broadcast address types, all 1's in the address is recognized by all nodes on a network, an 'All call' address contains the callsign only with the remaining fields set to all 1's and is recognized by all nodes with the same callsign, regardless of the USI.

There are two methods to uniquely address a node, by VPN or AX25 address. In the VPN mode, the USI is a 2-byte field composed from a unique identifier register within the node hardware, and in the AX25 compatibility mode the 5th byte is set to all 1's and the last byte contains the secondary station identifier (SSID), and 'C' and 'H' bits from the AX.25 protocol (denoted X). Table 23 illustrates the VPN coding.

Type	**Callsign Field**				**Unique Station Identifier**	
Byte	**1**	**2**	**3**	**4**	**5**	**6**
Broadcast	FF_{16}					
All call	Callsign **1-6** in Excess 40 format				FF_{16}	
VPN Address					VPN Address	
AX25 Compatibility					FF_{16}	$101XSSSS_2$

Table 23 VPN address coding

Callsign Compression

The callsign field characters are limited to the code set as illustrated in Table 24.

Character	**Coding**	**Character**	**Coding**	**Character**	**Coding**	**Character**	**Coding**
0	0	Space	10	J	20	T	30
1	1	A	11	K	21	U	31
2	2	B	12	L	22	V	32
3	3	C	13	M	23	W	33
4	4	D	14	N	24	X	34
5	5	E	15	O	25	Y	35
6	6	F	16	P	26	Z	36
7	7	G	17	Q	27	_	37
8	8	H	18	R	28	-	38
9	9	I	19	S	29	@	39

Table 24 Callsign character coding

The field shall be encoded using the following formula. Unused characters shall be replaced with a space (00).

$$F(3,0) = \sum_{n=1}^{n=6} C(n) + C(n-1) \times 40$$

Where F is the field value in bytes, C is the character and n the index into the callsign field. Each callsign can be up to 6 characters. The maximum value of a legally encoded callsign field is F423FFFF. A value of all 1's (FFFFFFFF) is a broadcast address to all stations, addresses outside these limits are not used.

The flag field contents for the headers symbol are listed in Table 25.

Bits	Field	Purpose
15-14	Fragmentation	Fragmentation status
13-10	Packet Coding	Packet data type. See Table 54.
9-8	Bandwidth	On air occupied bandwidth
7	FEC Present	Indicates FEC is present
6-4	Constellation	Constellation used
3	Hop table	Hop table present
2	Source Extended	Source address uses extension field
1	Destination Extended	Destination address uses extension field
0	Payload length MSB	MSB of payload length (Type 0 and 3 only)

Table 25 Frame Header Flag field definitions

Table 26 lists the value of the multi-bit fields.

Field Name	Value	Meaning
Fragmentation	0	Self-contained frame
	1	First frame of a segmented frame
	2	Middle frame
	3	Last frame
Bandwidth	0	NB
	1	ENB
	2	WB
	3	EWB
Constellation	0	QPSK/2FSK
	1	QAM-8/4FSK
	2	QAM-16/2GFSK
	3	QAM-32/4GFSK
	4	QAM-64
	5	QAM-128
	6	QAM-256
	7	QAM-512**Error! Bookmark not defined.**

Table 26 Flag field values

Hop and Multipoint Table

The hop table has two purposes, depending on the destination address. In a point to point frame, such as one derived from an AX.25 packet, it reflects the callsigns of up to 2 repeaters in the path to the destination. In a point to multipoint frame, it can contain up to four destination addresses. In the point to multipoint mode, up to four stations can be addressed at the same time.

Field	Size	Purpose
Repeater/Dest MAC Address	6	First Repeater or split frame address
Field 1 flags	1	Flags
Repeater/Dest MAC Address	6	Second Repeater or split frame address
Field 2 flags	1	Flags
Dest MAC Address	6	Third destination of a split frame address
Field 3 flags	1	Flags
Dest MAC Address	6	Fourth destination of a split frame address
Field 4 flags	1	Flags
LRC-8	1	LRC-8 Checksum
Total Bytes	29	

Table 27 Hop table contents

The flag field contents for the hop table are listed in Table 28.

Bits	Field	Purpose
7	Valid	This field is valid
6	HBR	In the p-p mode, this frame has been repeated
5	Multipoint frame	This frame is a multipoint frame
4	Unused	
3-0	Subcarrier map	Map of subcarriers allocated to this station

Table 28 Hop Table Flag field definitions

The size of the subcarriers mapping varies with the bandwidth occupied. Table 29 illustrates how the subcarriers are allocated to each destination callsign.

Mode	Carriers	Bit 3	Bit 2	Bit 1	Bit 0
NB	110	0-27	28-54	55-82	83-110
ENB	152	0-37	38-75	76-113	114-151
WB	240	0-59	60-119	120-179	180-239
EWB	324	0-80	81-161	162-241	242-323

Table 29 Symbol subcarrier mapping

In decoding a multipoint frame, the entire symbol is first decoded, then it is split into individual frames with the multipoint address used for the destination. The hop table is discarded. The data length is adjusted to the actual length.

Fragmentation

The fragmentation scheme limits the payload data size to 500 bytes, which enables a full-size UDP or TCP frame to be transmitted in three successive symbols. Each symbol contains the same header information, but a field in the header identifies the frame as one of three following types:

1. Single frame. A self-contained frame with a payload of less than 500 bytes.
2. Start frame. The first of a series of frames containing the first 500 payload bytes.
3. Middle frame. The second of a series of frames containing bytes 501 to 1000 of the payload.
4. Final frame. **Either** the last in a two-frame series containing up to the last 500 bytes, **or** the last in a series of three, containing bytes 1001 onward to a maximum of 1500.

Figure 7 illustrates the fragmentation scheme.

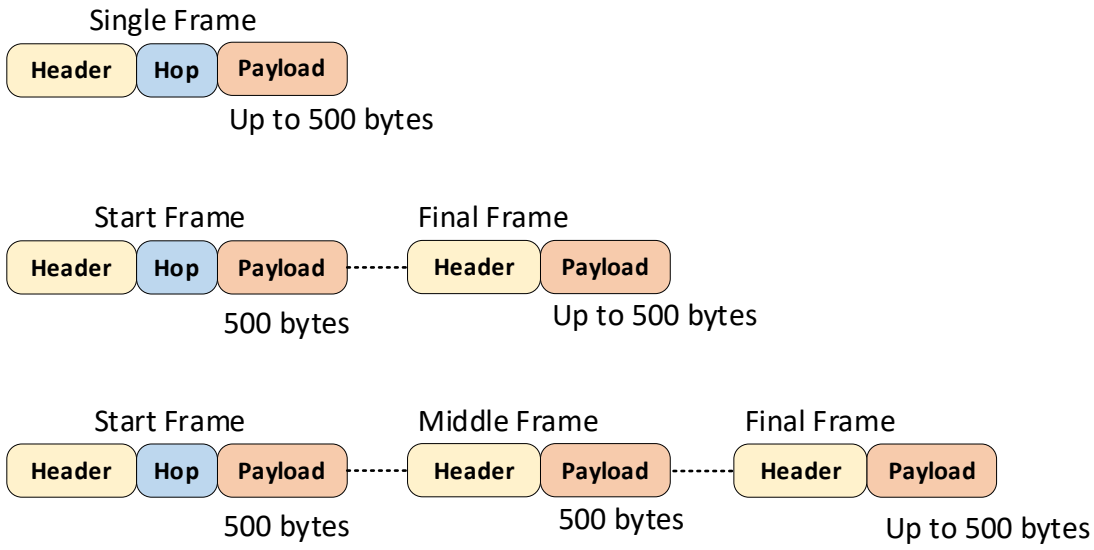


Figure 7 Frame Fragmentation Schemes

Buffering

Frames are packed into a modem transmit buffer that can hold 4095 bytes. As each frame is packed a timer is started, the transmission starts approx. 160ms after the last frame. This allows for maximum AX.25 packets to be sent back to back for optimum data throughput.

Platform Support

This system varies between different node types. The minimum functionality implemented contains radio specific parameters for each radio type, which are modified in the configuration menu.

Radio Parameters

Table 30 illustrates the radio setup parameters.

Parameter	Purpose	Value	Purpose
Mode	Specifies data mode	0-3	See Table 31.
Symbol Coding	Carrier Coding Scheme	0-7	See Table 32.
Frequency	Operating Frequency	n/a	Operating frequency[2]

Table 30 Radio Parameters

Table 31 illustrates the mode values.

Field Value	Standard
0	Narrowband
1	Enhanced Narrowband
2	Wideband
3	Ultra Wideband

Table 31 Data mod values for OFDM-AB standards

Table 32 defines the symbol coding.

Field Value	Bits	Constellation	Field Value	Bits	Constellation
0	2	QPSK	**4**	7	128 QAM
1	4	16 QAM	**5**	8	256 QAM
2	5	32 QAM	**6**	9	512 QAM
3	6	64 QAM	**7**	10	1024 QAM

Table 32 Data Symbols size definitions

[2] For documentation purposes only

Codec Configuration

The supernode architecture platforms use a TLV320AIC3014 codec from Texas Instruments as the primary digital to analog converter, which is configured through an I2C interface.

The basic specifications are shown in Table 33.

Parameter	Value	Units
Sample Rate	24	KHz
Number of Channels	1 or 2	
Word Size	16	bits
Signal to Noise Ratio	92	dB
Channel Separation	89	dB
Dynamic Range	91	dB
AGC Range	0-59.5	dB

Table 33 TLV320AIC3104 Code specifications

Clock Generator

An SI5351 clock generator creates up to three outputs from a single 27MHz frequency source, that has a stability specification of 0.5ppm. Table 34 lists the frequencies in use.

Output	Frequency	Usage
1	16MHZ	CPU and FPGA clock reference
2	24.576MHz	Codec 256x sample rate clock
3	21MHz	External modem reference clock

Table 34 Clock generator outputs

Modems and Physical I/O

This system contains the software support for each of the modem types, including any signal processing required for software defined modems.

Transceiver Architecture

Figure 8 illustrates the architecture of the OFDM modem and stack.

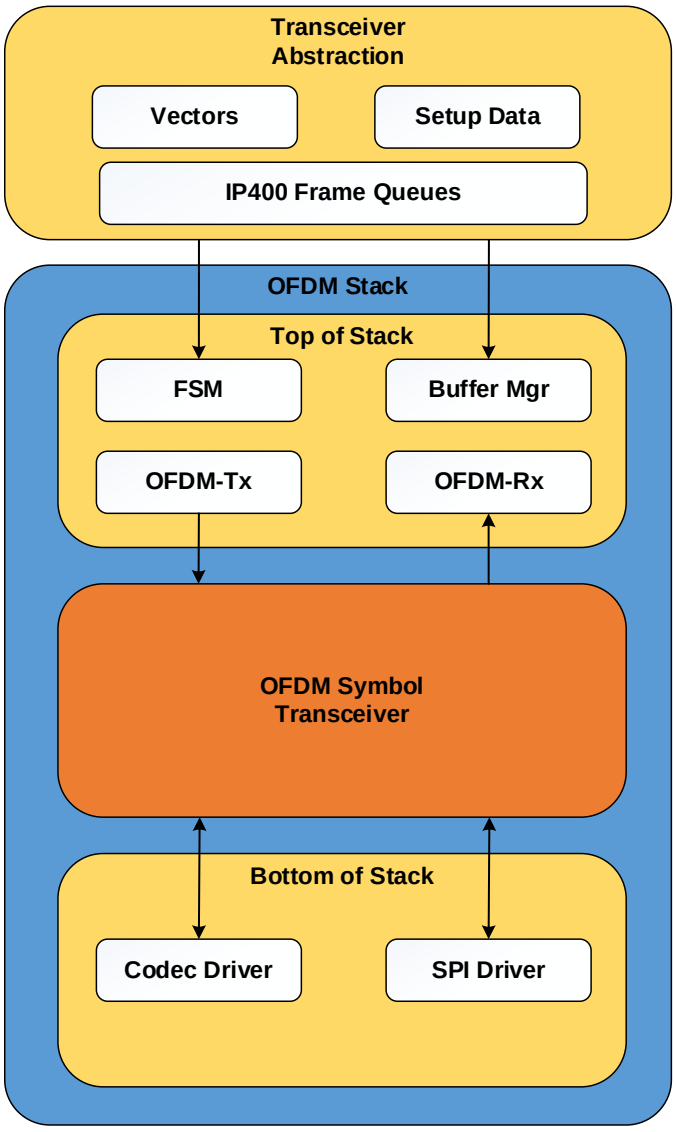


Figure 8 OFDM Transceiver Stack

Transceiver Abstraction

At the top is a module that enables different transceivers to be supported by the firmware. This module contains the default setup data for the modem type, critical vectors to access processing methods within each transceiver module, and queues for transmit and receive frames.

The AMNC module can support up the three different modems, the default is the built-in OFDM-AB modem, but provision has been made in the hardware to add another two, yet to be defined.

OFDM Stack

The OFDM Stack consists of three layers:

1. **Top of Stack.** This module contains a state machine to manage the receiver and transmitter, and the transitions. It also contains a buffer manager that converts IP400 frames into a series of bytes, and vice versa, thus decoupling the internal frame architecture from the modem. The transmit and module in this layer is responsible for converting bytes into symbols and creating the sync symbol and a CRC, and the receive module contains a mini state machine to look for synchronization, decode symbols and verifying the CRC.
2. **OFDM Symbol Transceiver.** This module contains the symbol coder and decoder, and also the sync detection algorithm, as well as the equalizer, which removes the channel perturbations and restores the original constellation. The bandwidth and constellation utilized are taken from the setup data in transmit, and this information is embedded in the sync symbol on receive, enabling all modes to be received without any additional setup parameters.
3. **Bottom of Stack.** This module sends and receives the OFDM symbols to and from the stack to the physical I/O device. Currently this can be a codec with an I2S bus, or an FPGA that connects to a codec.

Status Monitoring

Table 35 illustrates the usage of the four status LEDs.

LED	Colour	Indication
Power Good	Green	Power supplies are working correctly
HB	Green/Blinking	Software is operating properly
COS	Green	Receiver is enabled
	Red	Input level is too high
PTT	Green	Transmitter is active
	Red	FPGA is loading, being updated, or FPGA load failure

Table 35 LED Usage by platform

Modem Timing

Table 36 Illustrates the modem timing, with a CPU frequency of 200MHz, and code optimized for speed.

Component	Function	Timing (ms)	Margin
Sampling	Time to sample	45	
Transmitter	Sync Encode	12.5	32.5
	Symbol Encode, QAM-32 constellation	5.6	39.4
Receiver	Sync search	24	21
	Equalizer Calculation	19.6	1.4[3]
	Symbol Decode	18.8	26.2

Table 36 OFDM-AB Modem timing

[3] Equalizer calculation follows a successful sync

OFDM Transmissions

On-air transmissions are sent in the format of a superframe, which can contain up to 16K bytes of data.

Superframe format

The superframe is composed of three different fields:

1. Leader. A two-tone leader that is used to adjust the Automatic Gain Control (AGC).
2. Frame. From one to four data frames, each up to 4K in size.
3. Trailer. A period of silence at the end of the superframe.

Figure 9 illustrates the superframe format.

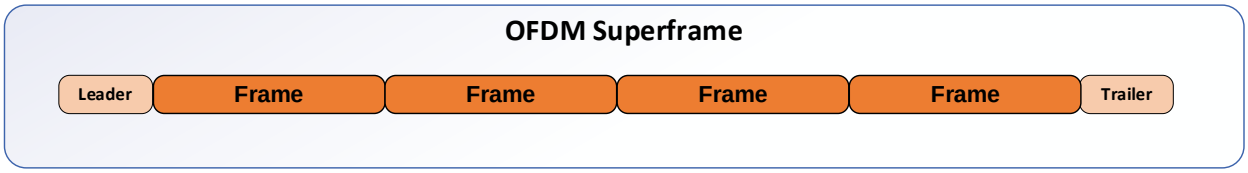


Figure 9 Superframe format

Data Frame

Each data frame consists of three fields:

1. Sync. A synchronization code that conforms to the Schmidl-Cox criteria.
2. Payload. Up to 4K of IP400 data frames, including any forward error correction.
3. CRC-16. A CCITT CRC-16 to validate the frame and FEC data.

Figure 10 illustrates the data frame format.

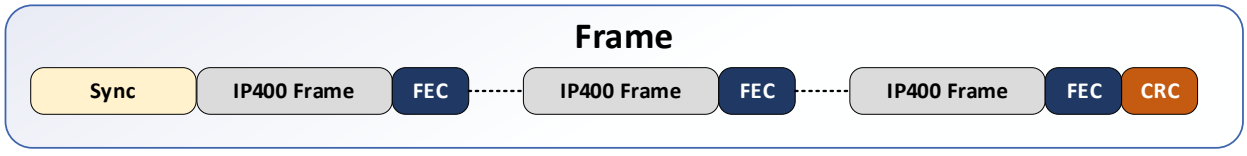


Figure 10 Data Frame Format

IP400 Frame

Each IP400 frame can contain up to 558 data bytes, which consist of a header, hop/multipoint table HMPT), and data fields (see Table 22).

Figure 11 illustrates the IP400 frame format.

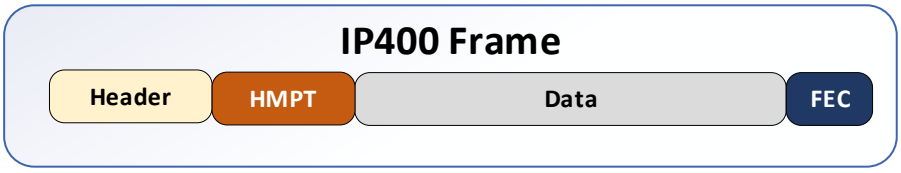


Figure 11 IP400 Frame format

Sync Code

The sync code is made up of two components: first, a PRBS (Pseudo-random binary sequence) of 32 bits in length is calculated, then it is modulated into a BPSK bit stream, and a data field is applied to the resulting stream, but inverting the phase of a one bit, and leaving it for a zero bit.

Field	Bits	Usage
CRC8	8	An 8-bit CRC to validate the data
FEC Code	1	An indication that the payload data contains an FEC code
Bandwidth	2	Data bandwidth, see Table 5.
Randomizer	2	Index of which randomizer was used in the data frame
Constellation	3	Constellation used to code the data, see Table 6.
Frame length	12	Length of the data frame

Table 37. Sync code fields

Randomizer

The randomizer uses a polynomial to generate a bitstream which is applied to the data to reduce the possibility of a sync code being detected in the data portion of the frame. One of four randomizers can be applied, Table 38 lists the polynomials.

Index	Polynomial
0	x^13+x^4+x^3+x+1
1	x^13+x^5+x^2+x+1
2	x^13+x^5+x^4+x^2+1
3	x^13+x^6+x^4+x+1

Table 38 Randomizer Polynomials

FPGA Digital Audio Ports

The digital audio ports are contained in the FPGA, which has four interfaces:

1. An SPI slave port for Audio PCM16 at 8Kb/s for the raspberry Pi.
2. A GPIO port for COR and PTT and an external modem.
3. An I2S slave that connects to the TLV320 codec.
4. An interface to the STM32F722 processor that has an SPI slave and GPIO.

Figure 12 illustrates the FPGA modules.

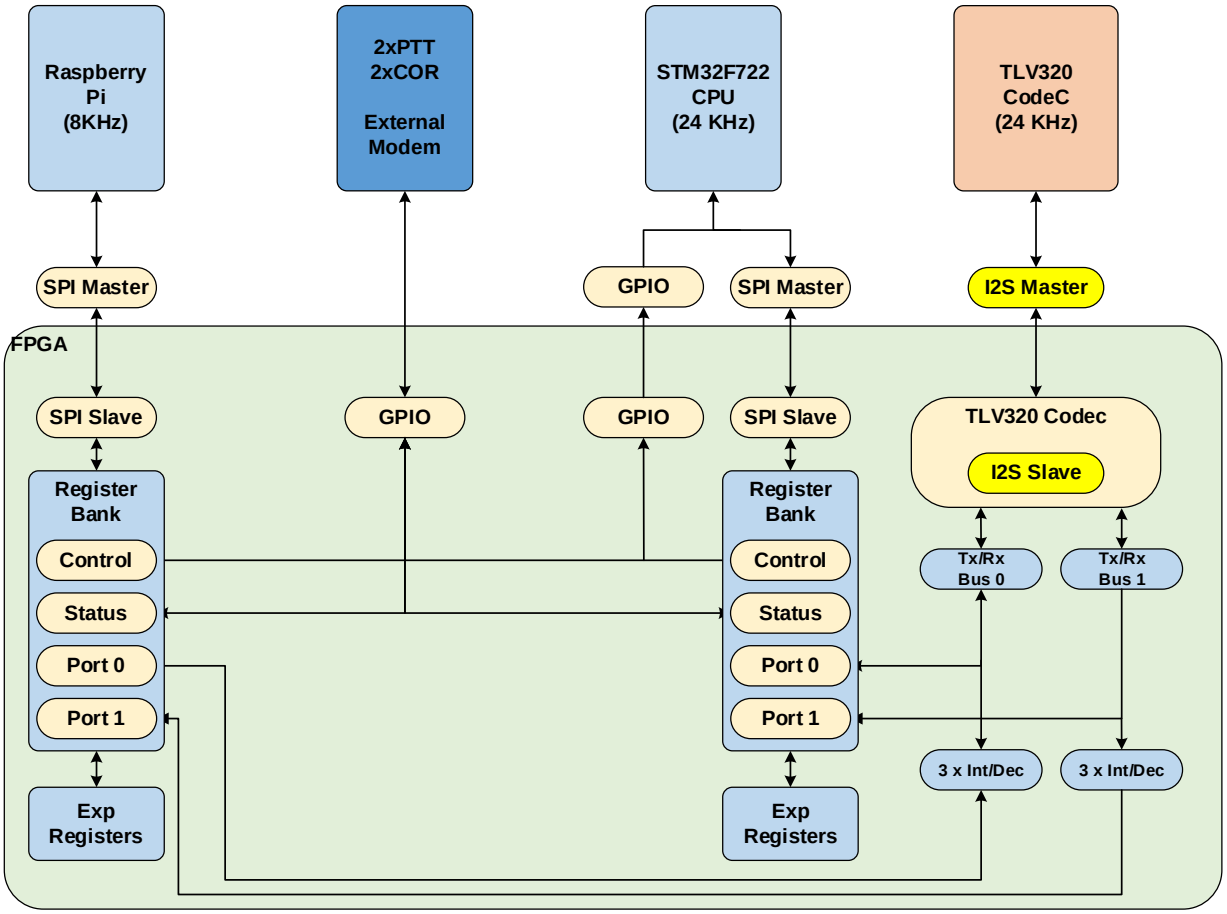


Figure 12 FPGA Digital Audio Ports

STM32F722 Interfaces

The STM32F722 interface consists of an SPI slave and a set of GPIO pins. The Cube MX configuration for the SPI is shown in Table 39 and the GPIO pins are listed in

Parameter	Value
Frame Format	Motorola
Data Size	16 bits
First Bit	MSB First
Prescaler	64 (1.640625 MHz)
Clock Polarity	Low
Clock Phase	2 Edge
CRC Calculation	Disabled
NSS Signal Type	Software

Table 39 OFDM-AB SPI Interface

SPI Interface

The timing diagram for the SPI interface is shown in Figure 13. Data from the CPU (MOSI) is sampled at mid-bit time and latched on the 16 clock transition. Data to the CPU (MISO) is latched on the rising edge of the clock, to be valid for the CPU to sample at mid-bit time.

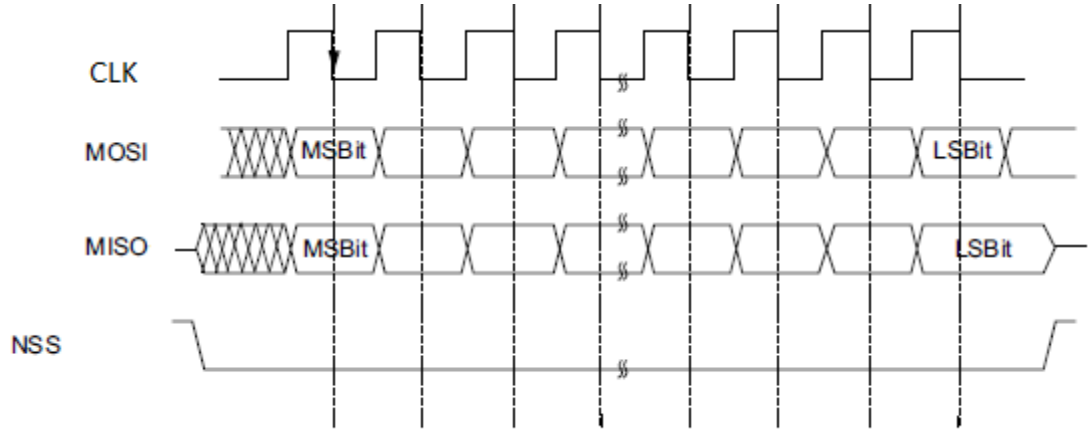


Figure 13 SPI Interface timing

The slave select signal (NSS), is controlled by software. The SPI is deselected when it is high, and active when it is low. Immediately after the assertion of NSS, the first word of each SPI transfer is a control word, which specifies the target register address, and read and write enables.

Control Word

Table 40 illustrates the bits of the control word.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Not Used								X	Reg Address			B	B	WE	RE

Table 40 SPI Control word

Where:

Field	Usage
B	Bank Address
Reg Address	Target register address
WE	Enables a write
RE	Enables a read

After the control word has been sent, the SPI interface enters a data mode, and remains in this mode until NSS is deasserted. This enables data symbols to be sent to the SPI using multiple DMA transfers, which do not have to be consecutive.

Bank Addressing

The CPU SPI has only one bank implemented, which are the FIFO registers. The other bank addresses are not implemented.

Bank Address	Module
00	FIFO registers
01	Not used
10	
11	

Table 41 CPU SPI Bank addresses

FIFO Registers

There are two FIFO register banks, one for each Codec port. There is only one implemented, which responds to bank address 00 and register addresses 0 through 7. Table 42 illustrates these registers. Two are reserved for an expansion bus, which can address up to 256 8-bit registers.

Radio Port	Reg Address	Read	Write
1	0	Receive Digital Audio	Transmit Digital Audio
1	1	FIFO Status (Tx[15:8], Rx[7:0])	FIFO Control (Tx[15:8], Rx[7:0])
1	2	Expansion Bus Address Register	Expansion Bus Data Register
1	3	PTT and Interrupt Status Register	PTT and Interrupt Control Register
2	4	Receive Digital Audio	Transmit Digital Audio
2	5	FIFO Status (Tx[15:8], Rx[7:0])	FIFO Control (Tx[15:8], Rx[7:0])
2	6	-	-
2	7	PTT Status Register	PTT Control Register

Table 42 OFDM-AB SPI Control and Status registers

The following tables illustrate the read and write contents of the registers and their meaning.

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	16-bit Sample															
1	MT	FL	LW	Words used < 32					MT	FL	HW	Words used < 32				
2		-							Expansion Read Data[7:0]							
3	IE	-		LB	TW	TE	RW	RE	CO	-						PTT

Table 43 Read Register Values

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	16-bit Sample															
1	Clr	RD	WR	-					Clr	RD	WR	-				TM
2	Expansion Address [7:0]								Expansion write Data [7:0]							
3	IE	-	LB	GP	TW	TE	RW	RE	CD	PTT1	CP	PTT2	PTT KEY			

Table 44 Write Register values

Expansion Bus

The expansion bus is only accessible through the first SPI register bank, and only on expansion module currently implemented, which contains the FPGA build number, and is illustrated in Table 45.

Bank	Address	Read	Write
CPU	0	FPGA code build number MSW	n/a
	1	FPGA code build number LSW	n/a

Table 45 Expansion Registers

The abbreviations are listed in Table 46.

Abbreviation	R/W	Description
MT	RO	Empty Flag. The FIFO is empty.
FL	RO	Full Flag. The FIFO is full, can be cleared by a reset
LW	RO	Low water mark. The FIFO holds less than 256 words.
HW	RO	High water mark. The FIFO holds more than 256 words.
RST	RW	Reset
SE	RW	Serial enable. Enables data in an out of AD73311
CO	RO	Carrier state
CD	WO	Enable carrier detect input
CP	WO	Carrier polarity, set to for positive, clear for negative
PTT	RO	PTT is on
PTT1, PTT2	WO	PTT Enable. Any value but must be exclusive nor of each other.
PTT KEY	WO	Must be '0110' to enable PTT
GP	RO	GPIO Bit from CPU
LB	RW	Digital Loopback enabled
IE*	RW	Interrupt enable on write; active on read
TW*	RW	Enables low water mark interrupt. See interrupt description.
TE*	RW	Enables transmit FIFO empty interrupt.
RW*	RW	Enables high water mark interrupt. See interrupt description
RE*	RW	Enables receive FIFO empty interrupt.
CE	RW	Enables carrier change of state interrupt

Table 46 Register Abbreviations

A (*) indicates an enable on write, and a status bit on read. The status bit indicates that the signal is active.

Carrier Detection

Carrier detection can be configured using the CD (Carrier detect) and CP (Carrier polarity) bits in the control registers. The values are shown in Table 47.

CD	CP	Source	Detection Type
0	**n/a**	Internal	Carrier detection is bypassed and is always active.
1	**0**	External	Active when COR input is positive, ground is inactive.
1	**1**		Active when COR input s at ground, inactive when input is positive

Table 47 Carrier detection configuration

Receiving Data

The receiver is enabled by writing the read and write enable bits in the FIFO control register, as well as the high water mark interrupt. When 512 samples are in the buffer, they can be transferred by DMA to the processor. The process repeats each time the HWM interrupt occurs. Receive data will not be written to the FIFO unless the carrier is detected, as described above.

Interrupt

The interrupt indicates a change of state in the FIFO. It has two transmit and two receive interrupts, which are gated by the interrupt enable, which has to be set for an interrupt to occur. Table 48 illustrates the FIFO interrupts. All interrupts are connected to the FPGA_IRQ pin.

System	Interrupt	Set	Cleared
Tx	TW	When the FIFO data count crosses the 256 to 255 boundary.	A write to the FIFO or by clearing the enable bit.
	TE	When the FIFO data count becomes zero.	A write to the FIFO or by clearing the enable bit
Rx	RP	When the FIFO data count crosses the 511 to 512 boundary.	A read from the FIFO or by clearing the enable bit.
	RE	When the FIFO is empty	A write to the FIFO by the codec, or by clearing the enable bit

Table 48 FIFO change of state interrupts

GPIO Pins

There are also seven GPIO pins connected to the FPGA, as illustrated in Table 49.

GPIO Port	FPGA Connection	Purpose
PA4	SPI_NSS	Slave select for SPI interface
PB0	FPGA_IRQ	Low edge sensitive interrupt input to the CPU
PB1	FPGA_SYSRESET	Reset input to FPGA modules
PC13	FPGA_CRESET	FPGA configuration reset
PC15	CDONE	Configuration done input
PC4	FPGA_GPIO1	Uncommitted GPIO for future use
PA8	FPGA_GPIO2	

Table 49 STM32F722 FPGA GPIO usage

Transmitting Data

Figure 14 illustrates the transmitter state machine.

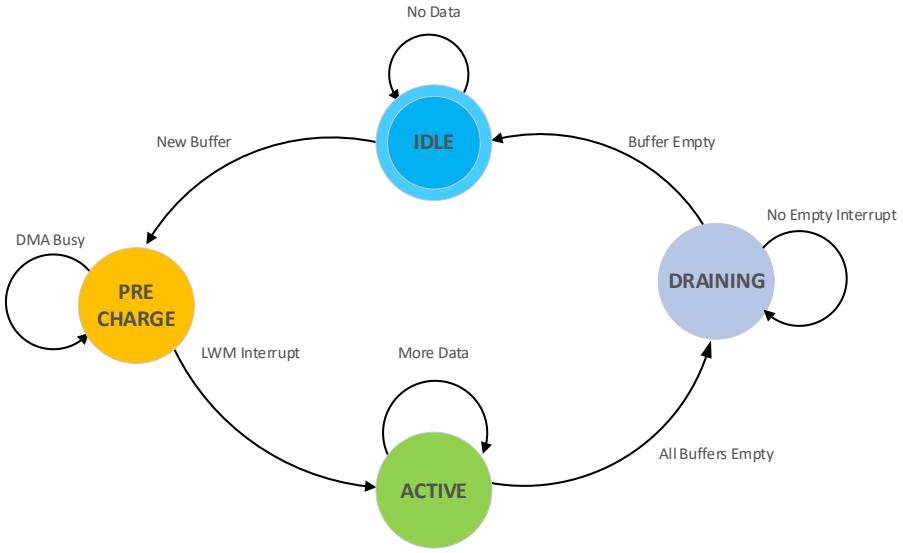


Figure 14 Transmitter state machine

There are four state in the transmitter state machine:

1. **IDLE.** The state machine starts up in this state, and when a new transmission starts, the WR enable bit is set in the control register, and a DMA transfer is initiated to the FIFO. The state is changed to PRECHARGE.
2. **PRECHARGE.** This state is entered when the buffer is being transferred out to the FIFO using DMA. Once complete, the RD enable is set, along with the LWM (low water mark) interrupt, and state is changed to ACTIVE. The FIFO will not spool out any data unitless the PTT has been previously asserted. This process adds about 10ms to the transmitter time and can overlap the transmit turn-on delay which can be up to 100ms.
3. **ACTIVE.** State to send all remaining data. It continuously refills the FIFO when the low water mark interrupt occurs, and when there is no more data to send, it enables the MT interrupt and enters the DRAINING state.
4. **DRAINING.** The transmitter remains in this state until the MT interrupt occurs, then it disables all interrupts, and then returns to the IDLE state.

Pi SPI Interfaces

There are two Pi SPI interfaces, as illustrated in Figure 15.

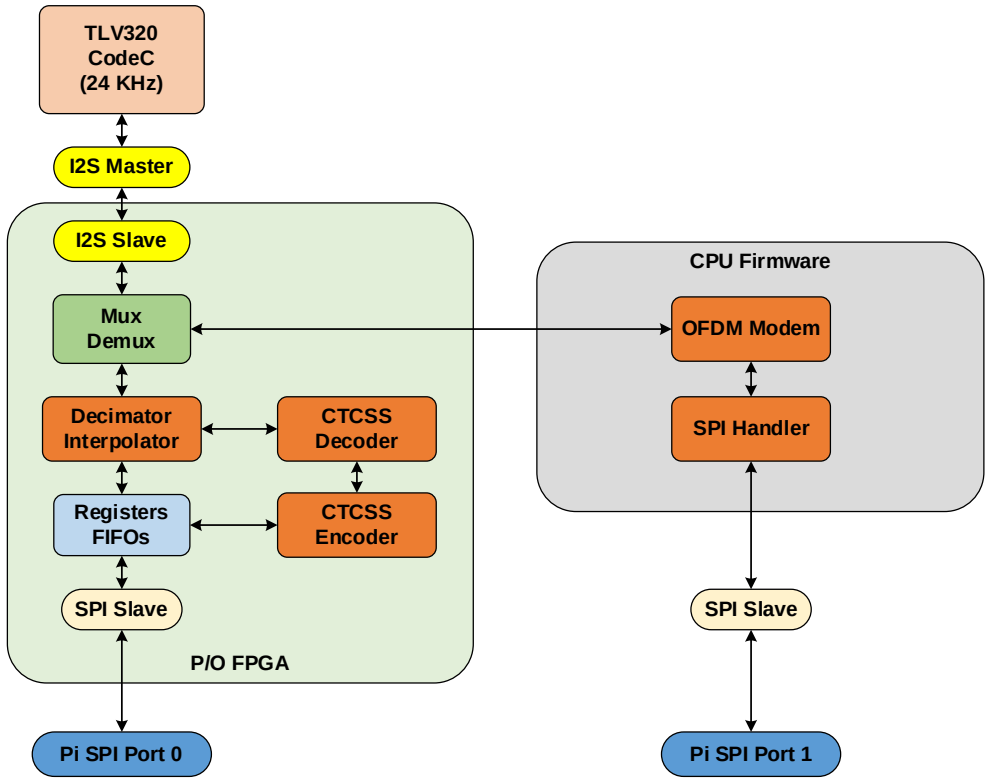


Figure 15 Pi SPI Interfaces

Pi SPI Port 0

SPI 0 is a 16-bit interface for PCM digital audio at 8 KHz. It has the same register set as the microcontroller but has no interrupt mechanism. The low and high water marks are also adjusted to 320 data samples, which represents 20ms of digital audio. This port also has access to the COR status and PTT controls. When transmitting, access to the codec by the modem is blocked.

Bank Addressing

The Pi SPI has up to three banks implemented, which are the FIFO registers, CTCSS encoder and decoder.

Bank Address	Module
00	FIFO registers
01	CTCSS Encoder/Decoder
10	Not used
11	

Table 50 CPU SPI Bank addresses

Pi SPI Port 1

Port 1 is an 8-bit data port that sends and receives IP400 frames that consist of a header and data fields. The SPI frame format is a slightly different from the on-air format, but follows the same format with an address field, hop table and data. The maximum data length is 500 bytes, but allowances have been made to accommodate an full length IP packet of 1500 bytes with fragmentation.

SPI 1 Frame Format

Table 51 describes the SPI frame format, which is a fixed-length bidirectional exchange of 524 bytes.

Field	Contents	Size (bytes)
Eye	IP4C	4
Status	Status	1
Offset Hi	High Byte of offset	2
Offset Lo	Lo Byte of offset	
Length Hi	Hi Byte of length	2
Length Lo	Lo Byte of length	
From Call	From Call	4
From VPN	From VPN	2
To Call	To Call	4
To VPN	To VPN	2
Coding	Coding	1
hop Count	Flags MSB	1
Flags	Flags LSB	1
Payload	Payload data	500
	Size in bytes	524

Table 51 SPI Frame Header format

Table 52 identifies the flag fields in the header. The MSB byte contains information about the packet coding and target modem, the LSB byte has channel coding information as well as callsign extensions.

Frame Flags	Size (bits)	Purpose	IP400 Frame field	SPI Header field
Destination Modem	2		Flags MSB	Status Byte
FEC Method	3			Coding byte
Constellation Type	3		Flags LSB	Flags MSB
Bandwidth	2			
Vacant	2			
Hoptable	1			Flags LSB
srcExt	1			
destExt	1			
payloadMSB	1			

Table 52 Frame flags definition

Table 53 defines the fragmentation field values.

Value	Type	Length	Payload content
0	Self contained	<500	Complete frame
1	First fragment	500	First 500 bytes of fragmented frame
2	Middle fragment	500-999	Second or last fragment, up to 500 bytes
3	Last fragment	1000-1499	Final fragment

Table 53 Fragmentation field values

Table 54 defines the packet coding values.

Value	Type	Purpose
0	UTF8_TEXT_PACKET	Used by internal chat application
1	DATA_PACKET	Download data for firmware upgrade
2	BEACON_PACKET	Beacon packet
3	AX_25_PACKET	AX.25 encapsulated packet
4	IP_ENCAPSULATED	TCP/UDP encapsulated packet
5	ECHO_REQUEST	Echo request to remote station
6	ECHO_RESPONSE	Echo response from remote station
7	LOCAL_COMMAND	Request for routing table lookup
8-15	USER_DEFINED	Available for user use

Table 54 Packet coding

Table 55 lists the modem qualifier fields and meaning.

Field	00	01	10	11
Modem	4FSK Primary	4FSK Secondary	OFDM-AB	Reserved
FEC Method	None	CCITT CRC-16	Trellis Coding	Reserved
Bandwidth	NB	ENB	WB	EWB

Table 55 Modem qualifier fields

SPI to IP400 Header mapping

Table 56 illustrates how SPI frames are mapped into an IP400 Frame header.

SPI Source Data			**Destination**	
Field	**Subfield**	**Size**	**Frame Header**	**Size**
Header (23 bytes)	Eye	4	n/a	
	spiStat	Lower 4 bits	Flags: fragmentation	2 bits
	Offset_hi	1	n/a	
	Offset_lo	1	n/a	
	Length_hi	1	Flags: payloadMSB	1 bit
	Length_lo	1	Length	1 byte
	From Call	4	Source MAC [0..3]	4 bytes
	From VPN	2	Source MAC [4..5]	2 bytes
	To Call	4	Dest MAC [0..3]	4 bytes
	To VPN	2	Dest MAC [4..5]	2 bytes
	Coding	4 bits	Flags: coding	4 bits
	Hopcount	1 byte	n/a	
Flags (1 byte)	Fragmentation	2 bits	Flags: fragmentation	2 bits
	Hoptable	1 bit	Flags: hoptable	1 bit
	SrcExt	1 bit	Flags: srcExt	1 bit
	DestExt	1 bit	Flags: destExt	1 bit
	Command	1 bit	n/a	
	noConnect	1 bit	n/a	
	Repeat	1 bit	n/a	
Ext Src Call	Callsign	4 bytes	srcExt	4 bytes
Ext Dest Call	Callsign	4 bytes	destExt	4 bytes
Hop Table	Address (2)	6-12 bytes	MAC [0..5]	6-12 bytes
(14 bytes)	Flags(2)	2 bytes	MAC [0..5]	2 bytes
Payload	Data	500-(Ext Src)-(Ext Dest)-(Hop Table)		

Table 56 SPI to IP400 Header Mapping

CTCSS Encoder and Decoder

The Continuous tone coded squelch system module provides a squelch detection to the audio path, and adds a tone to the transmit audio path.

Bank Address	Reg Address	Read (16 bits)	Write (16 bits)
1	0	Status Register	Control Register
1	1	Tx Divider MSW	
1	2	Tx Divider LSW	
1	3	Rx Coefficient	
	4	Tone Generator Gain	
	5	n/a	
2	6	Last Detection Peak MSW	Detect Threshold MSW
2	7	Last Detection Peak LSW	Detect Threshold LSW

Table 57 CTCSS encode/decoder register definitions

Address	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
10 Wr	Ena								ME						Tx	Rx
10 Rd	DET															

Table 58 Control/Status register definitions

The bit definitions are illustrated in Table 59.

Bit	R/W	Definition
Tx	W	Enable transmit tone
Rx	W	Enable Rx tone decoder
ME	W	Mute enable: Enables tone detection to mute modem audio path
Ena	W	Global enable for both transmit and receive
DET	RO	A tone has been detected

Table 59 Register Bit definitions

The tone generator divider values are illustrated in Table 60, and the tone detector coefficients are listed in Table 61.

ID	Tone	Divider	ID	Tone	Divider
YB	88.5	02D4FD	5A	156.7	0503AF
ZZ	91.5	02ED91	5B	162.2	0530BE
ZA	94.8	03089A	6Z	167.9	055F6F
ZB	97.4	031DE6	6A	173.8	058FC5
1Z	100.0	033333	6B	179.9	05C1BD
1A	103.5	034FDF	7Z	186.2	05F559
1B	107.2	036E2E	7A	192.8	062B6A
2Z	110.9	038C7E	M1	203.5	068312
2A	114.8	03AC71	8Z	206.5	069BA5
2B	118.8	03CD35	M2	210.7	06BE0D
3Z	123.0	03EF9D	M3	218.1	06FAAC
3A	127.3	0412D7	M4	225.7	0738EF
3B	131.8	0437B4	9Z	229.1	0754C9
4Z	136.5	045E35	M5	233.6	0779A6
4A	141.3	048587	M6	241.8	07BCD3
4B	146.2	04ADAB	M7	250.3	080275
5Z	151.4	04D844	0Z	254.1	082196

Table 60 CTCSS Tone encoder divider values

ID	Tone	Coefficient	ID	Tone	Coefficient
YB	88.5	7FA9	5A	156.7	7EFB
ZZ	91.5	7FA4	5B	162.2	7EE9
ZA	94.8	7F9D	6Z	167.9	7ED6
ZB	97.4	7F98	6A	173.8	7EC1
1Z	100.0	7F92	6B	179.9	7EAB
1A	103.5	7F8B	7Z	186.2	7E9D
1B	107.2	7F83	7A	192.8	7E93
2Z	110.9	7F7A	M1	203.5	7E85
2A	114.8	7F71	8Z	206.5	7E79
2B	118.8	7F67	M2	210.7	7E6A
3Z	123.0	7F5D	M3	218.1	7E5E
3A	127.3	7F52	M4	225.7	7E4E
3B	131.8	7F46	9Z	229.1	7E41
4Z	136.5	7F38	M5	233.6	7E2F
4A	141.3	7F2B	M6	241.8	7E0F
4B	146.2	7F1C	M7	250.3	7DEC
5Z	151.4	7F0C	0Z	254.1	7DDC

Table 61 Goertzel Detection algorithm coefficients

Operating the Software

Main Menu

The main menu is invoked by connecting to the UAR/T on the node with a serial connection at 115200bps. After connecting, the enter key will invoke the menu. It is illustrated in Figure 16.

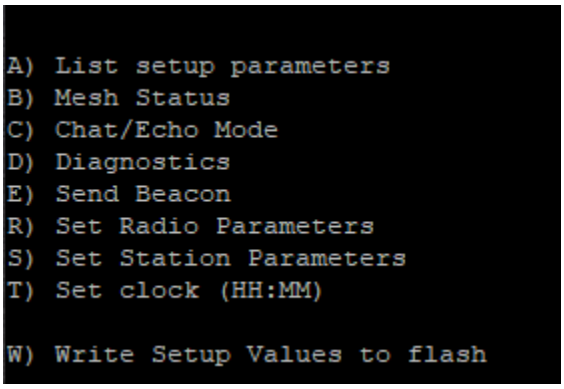


Figure 16 Main Menu

The menu items are described in Table 62.

Menu Item	Function
A	Lists the current setup parameters, including clock, station and radio
B	Displays the mesh table contents
C	Enters chat mode and echo modes
D	Enters the diagnostics menu
E	Sends a beacon frame immediately
R	Sets the radio parameters
S	Sets the station parameters
T	Sets the time of day clock
W	Writes the setup data to the flash

Table 62 Main Menu Items

Item W is used to write the setup parameters to the flash memory after modification. When powering up, if the memory is valid, the parameters are loaded. If not, then a default set of parameters is used, which can be modified accordingly.

It is possible that reloading the firmware may cause the parameters to revert to the default. This is due to changes in the setup memory layout.

Setup Parameters

A list of all the current setup parameters can be seen from the main menu item ‘A’. There are four groups of parameters:

1. Firmware version, system time, radio ID and VPN address.
2. Station callsign, AX.25 compatibility mode and description
3. Location data and capabilities
4. Repeat (digipeat) mode enable
5. Beacon interval

Figure 17 illustrates them.

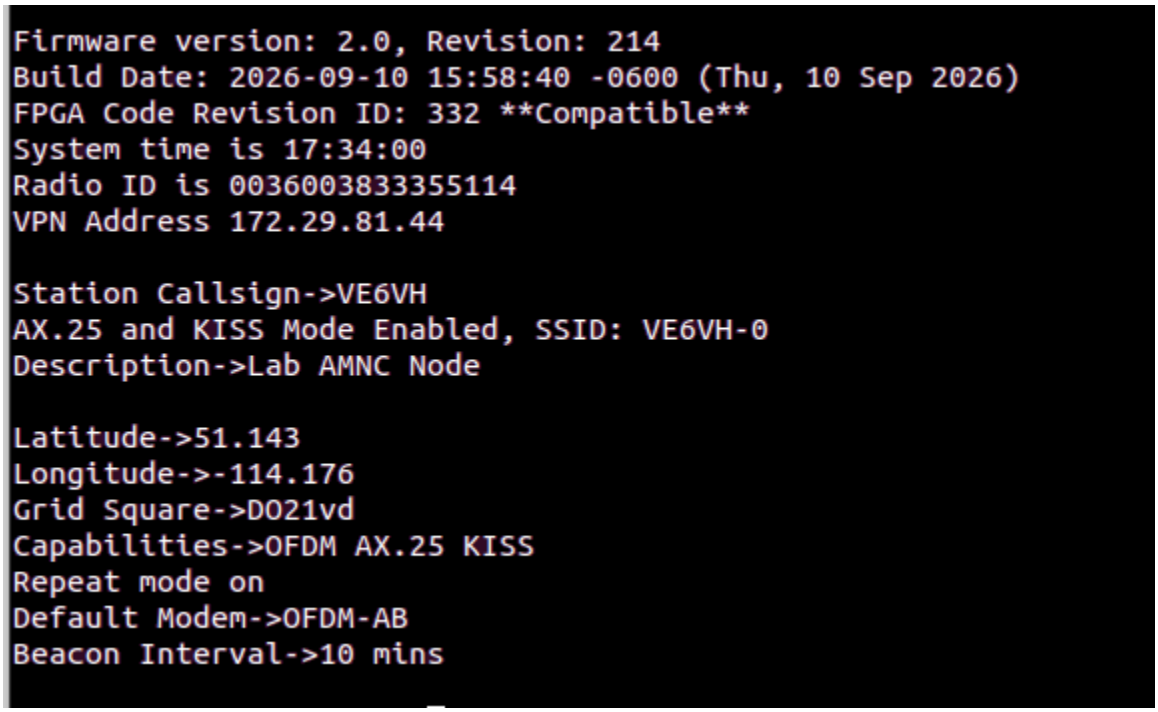


Figure 17 Setup Parameters

Mesh Table

Figure 18 illustrates the mesh table contents, and Table 63 explains them.

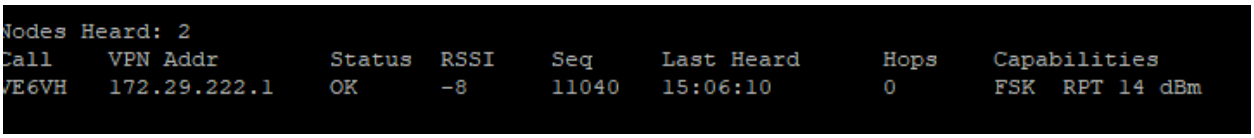


Figure 18 Mesh Table

Item	Explanation
Call	Callsign of the transmitting station. In AX.25 mode, also has the SSID
VPN Address	The network VPN address of the station
Status	OK if heard in the last hour, LOST if not.
RSSI	Receive signal strength or SNR when SYNC received (in dBm)
Next Seq	Next anticipated sequence number
Last Heard	TOD clock reading when last heard
Hops	Number of repeat hops, 0 indicates a direct signal
Capabilities	Data rate, repeat capabilities and transmitted signal strength

Table 63 Mesh Table explanation

The station capabilities can only be populated upon receipt of a beacon frame. If a non-beacon is received from a new station, it is added to the table without this field. If a beacon is heard subsequently, all the fields are updated. See Table 64 for details of these.

Bit	Capability	Meaning
0	FSK	Station can send 2 or 4 FSK
1	OFDM	Station is capable of OFDM transmissions
2	AX.25	Station has the AX.25 mode enabled
3	REPEAT	Station can act as a digipeater
4-7	EXT	SSID when in AX.25 compatibility mode

Table 64 Capabilities field

Chat/Echo Mode

The chat mode is an interactive exchange between two nodes. Frames can be broadcast to all nodes, to all nodes with the same callsign, or to a specific node.

When invoked, it defaults to the chat mode with a broadcast address. Control keys are used to change the chat status, as shown in Table 65.

Key	Purpose
ESC	Enables entering a new destination address.
CTRL+'R'	Toggles repeat mode
CTRL+'E'	Enters echo mode
CTRL+'D'	Toggles dump mode
CTRL+'Z'	Exits chat and returns to the main menu
ENTER	Sends the current frame/ends an entry
BACKSPACE	Deletes the last character entered, or more if repeated

Table 65 Chat mode contol keys

Sending a Chat Frame

First destination address must be set, by depressing the ESC key when the cursor is at the beginning of the line, from that point there are three options:

1. Depressing the ENTER key right away will set the address to broadcast, which reaches all nodes on the network.
2. Enter a callsign and depress the ENTER key. From there one of three things can happen:
    a. If the call is not in the mesh table, it will be rejected, and the last destination will remain unchanged.
    b. If that call is unique and, in the table, then the destination will be set to the MAC address of that node. Frames will only be sent to that particular one.
    c. If that call is not unique, the destination will be set initially to all nodes with that callsign, and a list of all the entries found is displayed. Each entry has a callsign followed by a dash ('-') and an index into the mesh table.
3. Enter callsign, dash and index. If found in the table, the destination will be set to that node only.

Chat frames are displayed as follows:

- Originator callsign and VPN address
- Destination callsign and VPN address
- Number of times the frame was repeated
- Sequence number of the frame
- Data message

Dump Mode

The dump mode changes the display format to hexadecimal and also includes the frame header information. Useful for debugging.

Echo Mode

The echo mode is designed to query a node to see if it can be reached, and whether it will respond. As with the chat mode, the destination address must be set first. The CTRL/E key is multi-functional as follows:

1. The first depression enters the interactive mode.
2. The second enters a timed mode.
3. The third returns to normal chat mode.

In the interactive mode. Depressing the ‘Enter’ key will send a query to the remote station. The response is displayed in the current chat mode.

In the timed mode, an echo request frame is sent to the remote station approximately once every 5 seconds.

Diagnostics Menu

Figure 19 illustrates the diagnostics menu and Figure 19 explains them.

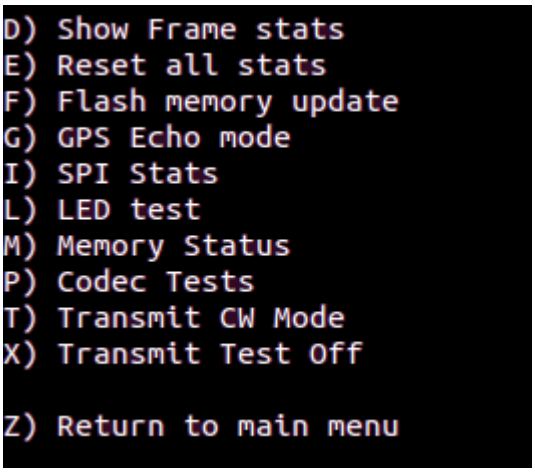


Figure 19 Diagnostics Menu

Menu Item	Function
D	Shows the frame statistics
E	Reset all statistics
F	Enables downloading the FPGA from the debug UART (requires programmer board)
G	Enables GPS Echo mode, if installed
I	Dumps the SPI statistics
L	Runs an LED test
M	Displays the memory status
P	Enters the codec tests menu
T	Transmits a CW signal
X	Turns of previous tests and returns to normal operation
Z	Returns to the main menu

Table 66 Diagnostics Menu

Frame Statistics

Figure 20 illustrates the statistics for the OFDM-AB modem.

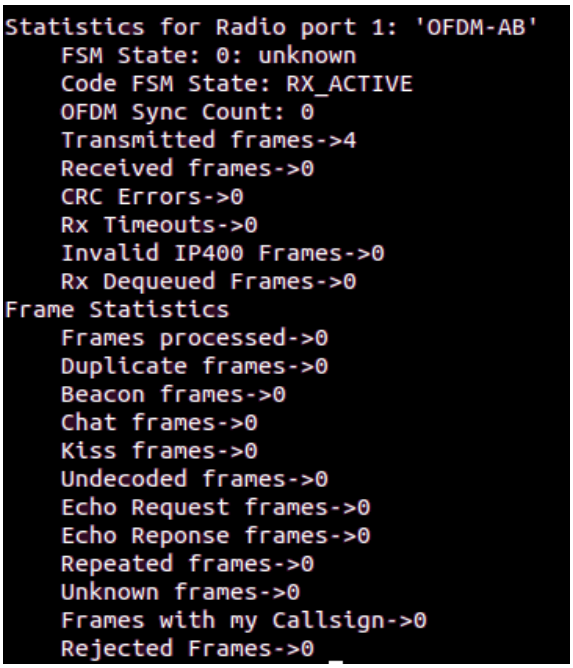


Figure 20 OFDM-AB Modem Statistics

Item	Explanation
FSM State	OFDM State machine
Code FSM State	State of the internal code state machine
Transmitted Frames	The number of transmitted frames since the last restart
Received Frames	Frames received with good CRC
CRC Errors	The number of frames received with FEC errors.
Rx Timeouts	The number of times the receiver timed out
Invalid IP400 Frames	Frames with an unrecognizable header
Rx Dequeued frames	Frames sent on for internal processing
Frame Statistics	Frame Processor statistics (see Table 68)

Table 67 Frame Statistics

Table 68 illustrates the frame processor statistics.

Item	Explanation
Frames Processed	Number of frames presented to the frame processor
Duplicated Frames	Frames rejected as being duplicate
Beacon Frames	Beacon frames processed
Chat Frames	Chat frames processed
KISS frames	KISS frames processed
Undecoded Frames	Frames not decoded by the node
Echo Request Frames	Echo Requests received
Echo Response frames	Echo responses sent
Repeated Frames	A count of the number of frames repeated
Unknown Frames	Frames that could not be recognized
Frames with my Callsign	Frames in which my callsign was found (in digipeat mode)

Table 68 Frame Processor Statistics

SPI Statistics

Table 69 illustrates the SPI statistics

Item	Explanation
SPI State	State of the SPI: active or not active
IP400->SPI Frames	Received frames sent to the SPI
SPI->IP400 Frames	Transmit frames received from the SPI
SPI Single Frames	Frames with a payload less that 500 bytes
First Fragment Frames	First 500 bytes of a larger frame
Last Fragment Frames	Remainder of a large frame
Discarded Frames	Frames without the correct ‘IP4C’ eye

Table 69 SPI Statistics

Memory Status

This menu item is for developers to determine if any memory leaks have occurred. Refer to the source code for calls to the allocator, and the Free RTOS manual for a description of the heap.

Codec Tests

Figure 21 illustrates the codec tests

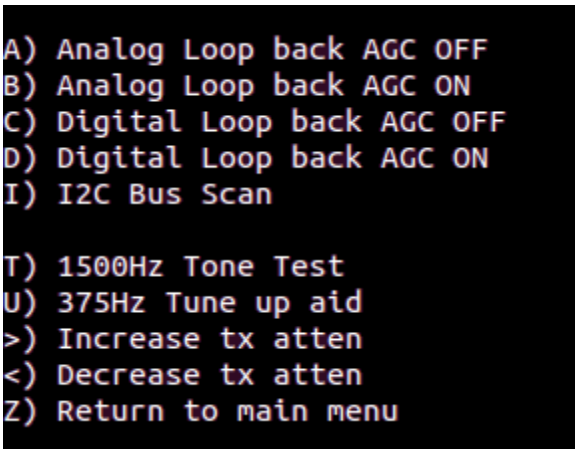


Figure 21 Codec Tests Menu

Item	Explanation
Analog Loop Back AGC Off	Enables a loopback at the A/D Interface with no AGC
Analog Loop Back AGC On	Enables a loopback at the A/D Interface with AGC enabled
Digital Loopback AGC Off	Enables loopback at the FPGA with AGC enabled
Digital Loopback AGC On	Enables loopback at the FPGA with AGC enabled
I2C Bus Scan	Scans the I2C bus devices for proper operation
1500Hz tone test	Keys the transmitter and sends out a 1500Hz tone
375Hz tune up aid	Keys the transmitter with a 375Hz tone
Increase tx atten	Increases the transmitter output attenuation.
Decrease tx atten	Decreases the transmitter output attenuation

Table 70 Codec test menu

Note: if the transmitter output attenuation is changed, it is not stored in the setup flash until an update is done from the main menu.

Send Beacon

This menu item sends a beacon frame immediately. The payload is a comma-delimited string, the contents are shown in

Field	Contents
Fix Source	Source of lat/long data, **FIX** for setup data, **GPS** from a receiver
Latitude	Current Latitude if GPS is connected, else supplied data.
Longitude	Current Longitude, same as above
Speed	Speed if in motion, null field if in FIX mode
Time	System or GPS time
Grid Square	Grid square from setup data
Description	System description from the setup data

Table 71 Beacon Frame Payload

LED Test

The LED test will cycle through the patterns shown in the following table:

Test	LED	Common use	Colour
1	LED 2	Receiver active	Green
2			Red
3	LED 3	Transmitter active	Green
4			Red
5	LED 1	Heartbeat	Green

Table 72 LED Test Cycling

GPS Echo Mode

The NMEA sentences from the GPS receiver are dumped to the screen as shown below:

```
GNVTG,160.14,T,,M,0.23,N,0.42,K,A*
GNGGA,040907.000,5108.6054,N,11410.5759,W,1,07,1.53,1275.4,M,-17.5,M,,*
GLGSA,A,3,86,85,71,,,,,,,,,,1.83,1.53,0.99*
GPGSV,4,1,14,18,73,294,36,29,50,156,22,23,42,220,26,15,40,130,29*
GPGSV,4,3,14,27,07,315,,20,05,057,,10,03,225,16,07,02,007,15*
GLGSV,3,1,11,86,75,007,33,70,53,059,,71,50,151,26,87,34,305,18*
GLGSV,3,2,11,85,32,101,22,77,21,285,,78,14,344,,72,09,183,*
GNRMC,040907.000,A,5108.6054,N,11410.5759,W,0.20,160.14,070225,,,A*
GPGSA,A,3,13,18,29,05,15,,,,,,,,1.74,1.42,0.99*
GNRMC,040903.000,A,5108.6057,N,11410.5757,W,0.35,160.14,070225,,,A*
GNVTG,160.14,T,,M,0.35,N,0.65,K,A*
GLGSA,A,3,86,85,71,,,,,,,,,,1.83,1.53,0.99*
GPGSV,4,1,14,18,73,294,36,29,50,156,22,23,42,220,26,15,40,130,29*
GPGSV,4,4,14,30,01,038,,45,,,*
GLGSV,3,1,11,86,75,007,33,70,53,059,,71,50,151,26,87,34,305,18*
GLGSV,3,3,11,76,07,243,,68,04,020,,69,01,019,*
GNRMC,040907.000,A,5108.6054,N,11410.5759,W,0.20,160.14,070225,,,A*
GPGSA,A,3,18,29,05,15,,,,,,,,,1.90,1.62,1.00*
GNRMC,040908.000,A,5108.6053,N,11410.5759,W,0.26,160.14,070225,,,A*
GNVTG,160.14,T,,M,0.20,N,0.36,K,A*
GLGSA,A,3,86,85,71,,,,,,,,,,1.74,1.42,0.99*
GNVTG,160.14,T,,M,0.35,N,0.65,K,A*
```

Figure 22 GPS NMEA sentences

While this mode is active, two keys are processed:

1. Pressing the ‘ESC’ key will pause and prompt for the enter key to be hit before returning to the main menu. The data remains on the screen.
2. Pressing the ENTER key will clear the screen and return to the main menu.

Setting the clock

The time of day is entered as HH: MM. The clock has a granularity of 10 seconds and uses 24 hour format, and knows nothing about time zones, as it considers you are in your home zone.

Radio Setup Menus

To radio setup menus can be invoked from the main menu item 'R'. Figure 23 illustrates the OFDM-AB radio setup menu. For the numerical mapping of some parameters, type the '?' key for help, if prompted.

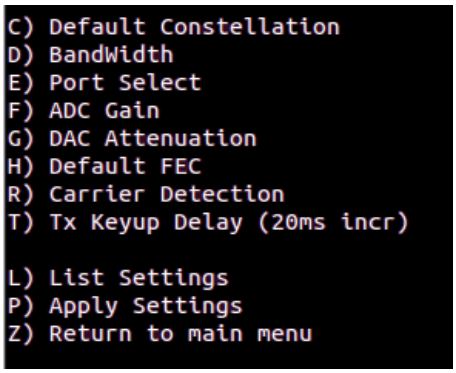


Figure 23 OFDM-AB Radio Setup Menu

Menu Item	Function
C	Sets the transmit default constellation. Table 6.
D	Sets the default transmit bandwidth. See Table 5.
E	Selects the port, 0 or 1.
F	Sets the ADC gain.
G	Sets the DAC Attenuation.
H	Sets the default forward error correction scheme
T	Sets the transmit keyup delay, in increments of 20ms.
R	Sets the carrier detection method
L	List the current settings. See Figure 24.
P	Applies setting changes immediately.
Z	Returns to the main menu.

Table 73 Radio Setup Meny Item

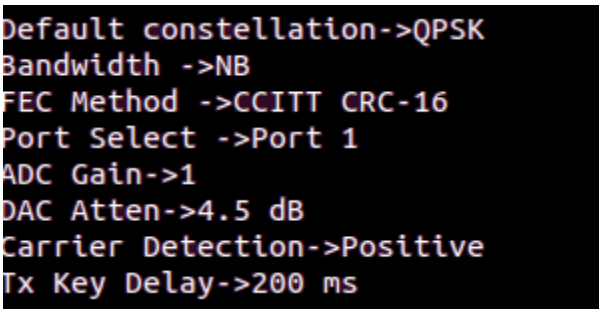


Figure 24 List of Radio Settings

Note that any of these setting are not permanent until written to flash from the main menu.

Station Setup Menu

The station setup menu is constant for all modes and is invoked from the main menu item 'S'. The menu is illustrated in Figure 25.

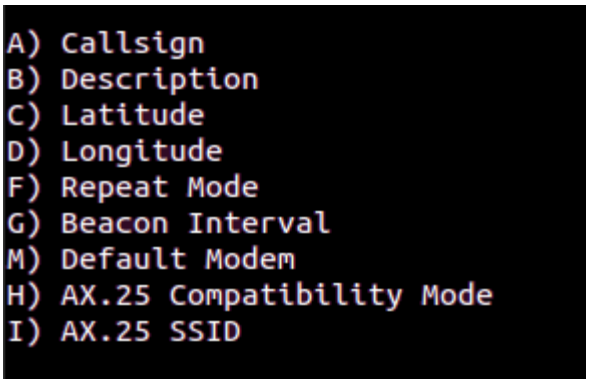


Figure 25 Station Setup Menu

Item	Purpose	Format
Callsign	Station callsign	Up to 6 characters[4]
Description	A description of the station	Up to 32 characters
Latitude	Latitude, positive for N, negative for S	±xxx.xxx[5]
Longitude	Longitude, positive for E, negative for W	±xxx.xxx
Repeat Mode	Set the repeat flag in sent frames	Y or N
Beacon Interval	Interval between beacons, normally 5 mins	xx minutes
Default Modem	Sets the default modem	0 to 2
AX.25 Compatibility	Enables the AX.25 addressing mode	Y or N
AX.25 SSID	The AX.25 SSID value	0 to 15

Table 74 Station setup parameters

[4] An extended method has been designed but not yet implemented
[5] Used in beacon frames, unless a GPS receiver is connected

Building the Code

All variants of the code are built using the ST Microelectronics CubeIDE platform, at version level 2.1.1 or later. There are three preliminary steps to building the code:

1. Unzip the project for the required platform and import into the IDE.
2. Import the common code.
3. Import the platform specific code.

Hardware Platform	Project zip file	Platform code	Object File	Flash Method
AMNC	AMNC.zip	AMNC/OFDM	SuperNode.elf	

Table 75 Code Components

Prior to building the code, add the source directories to the project first:

1. Right click on the project in the explorer window, choose ‘Properties.
2. From the dialog box, expand C/C++ General, then Paths and Symbols.
3. From the source location tab, click ‘Add Folder’.
4. Choose the common source folder and click ‘Apply’.
5. Repeat for the platform code as shown in Table 75.

Right click on the Project again, this time choose ‘Clean Project’, then ‘Build Project’.

Downloading

Table 75 also lists the flashing method for each of the hardware platforms. The debugger is contained in the CubeIDE platform, otherwise the object file can be copied to the target platform and flashed with the appropriate shell script.

Those platforms that do not support a debugger must use the flash shell script method.

Support Software

Remote Ethernet System

A raspberry Pi provides can provide a remote interface to the AMNC over ethernet, by connecting to the SPI and UAR/T ports, and a path for digital audio to be transmitter over an IP400 network.

Figure 26 illustrates the subsystem.

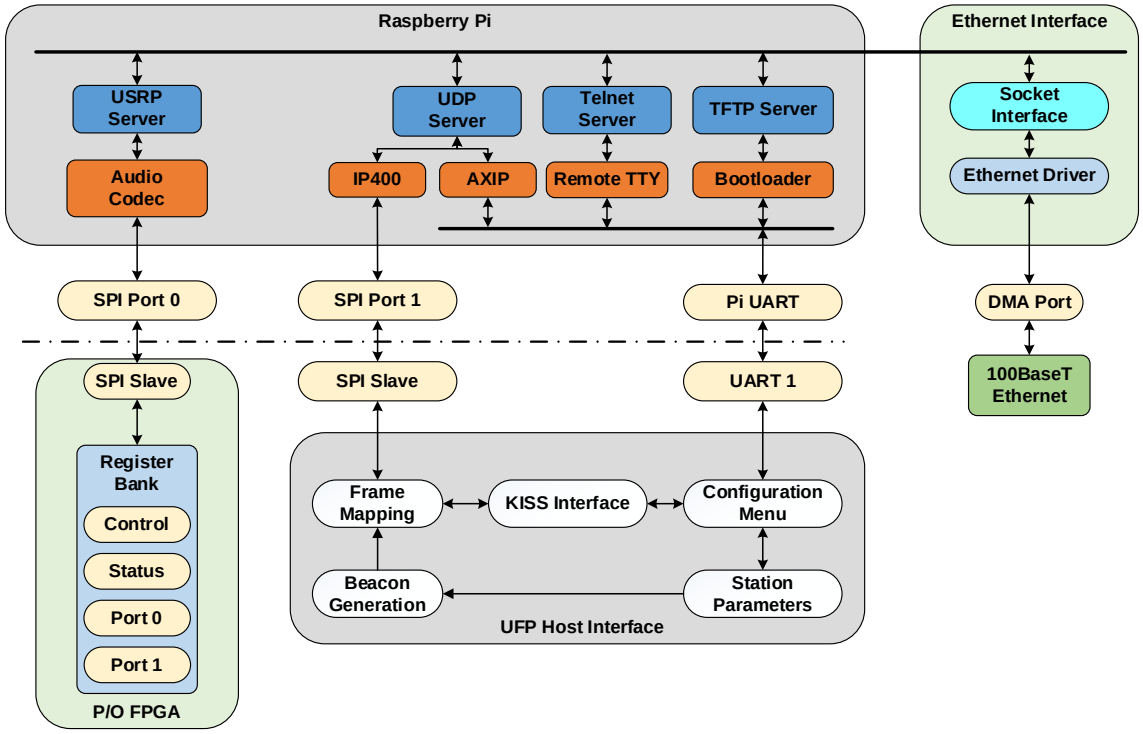


Figure 26 Ethernet Subsystem

There are four socket clients and corresponding applications that use ethernet:

1. A trivial file transfer server (TFTP) and bootloader that provides a facility to upgrade the system firmware.
2. A telnet interface that provides a connection to the host interface UART for configuration and setup data.
3. A UDP server that accepts either IP400 or AXIP packets. IP400 packets are routed to the SPI interface, and AXIP to the Kiss interface.
4. USRP server. A server that sends and receives USRP packets that contain digital audio.

FPGA Flash Programmer

The flash programmer module interfaces with the KISS mode module (See Table 15) to provide a facility to update the FPGA flash memory. There are two components, the KISS module has a unique command to start the upload process, and a host program downloads the FPGA hex file. There are two versions, one runs on a PC under windows, the second under Linux.

KISS mode module

A unique KISS command enters the boot loader. For a detailed description of the command,

The host to programmer protocol is shown in Figure 27.

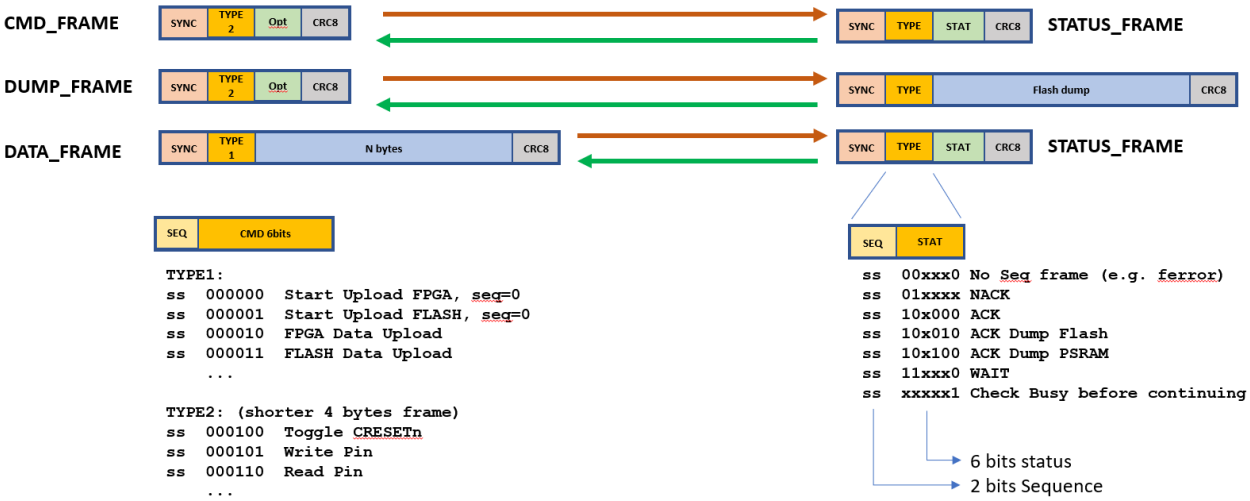


Figure 27 Flash programming protocol

The flash programmer has the following command line options:

Option	Purpose	Default
-c	TTY port or COM Port	Linux only: /dev/ttyAMA0.
		Windows specify com port number.
-b	Port baud rate.	115200
-k	Sends KISS command, use with menu UART only.	Off.
-f	Filename of hex file to download	download.hex
-t	Test mode. Sends numbered test frames	Off
-d	Debug mode. Dumps frame contents	Off

Table 76 FPGA programmer command line options

The kiss command can be omitted when using the debug UART, and the download initiated at the Diagnostics menu. When using the main UART, it must be specified to enter the download mode.

UDP to SPI software

The UDP to SPI software (ip400spi) has been developed to provide a mechanism to send and receive packets to the node using the IP user datagram (UDP) protocol from a raspberry pi. The software is invoked on the command line, using the parameters shown in Table 77.

Parameter	Argument	Meaning
-B	None	Sends a single test frame, built into the software in frame.c
-s	/dev/spiX.Y	Specifies the SPI port. Usually /dev/spi0.0
-n	Name or IP	Name or IP address of the host receiving the packet
-p	nnnn	Port address of the host
-m	nnnn	Sending port address
-d	x	Sets the debug level to x.
-l	n	Sets loopback mode. See below and Table 79.
-h	None	Prints the help message

Table 77 Pi utility software command parameters

The program is terminated by typing Control/C at the keyboard. It can also be run in the background, or as service.

The debug level is a bit field that has the following values:

Bit	Value	Debug
0	1	Enables logging of debug messages
1	2	Enables SPI debugging. Removes real time performance

Table 78 Pi Utility debug levels

Loopback mode

Loopback mode is a debug mode intended for debugging formatting of packets into SPI payloads. When active, the SPI is not enabled, instead a transmitted packet goes through the motions of being fragmented, but instead of being sent over the SPI, it is looped back as if it was received.

This exercises both the transmit and receive fragmentation modes, a successful test methodology should be to use the packet generator to generate packets, which are sent to a PC running Wireshark using the loopback mode. The PC should be running the dissector.

Value	
1	SPI loopback. Simulates an SPI transmit/receive cycle by copying data from Tx to Rx
2	IP400 Frame loopback. Exercises the node SPI code and loops back at frame level
All others	No loopback operation performed

Table 79 Loopback mode arguments

UDP Packet Generator

The UDP generator software (UDPGen) generates a series of packets that can be used as test cases for the UDP to SPI software. Table 80 lists the command line parameters.

Parameter	Argument	Meaning
-n	Hostname	Name or IP address of the host receiving the frame
-p	Host Port	Port number on the host
-m	Orig Port	Originating port number
-f n	Frame type	Frame type to send (see Table 81)
-r	Repeat count	Number of times to send the frame (default 1)
-D	Debug mode	Enables debug messages to be displayed
-h	Help	Prints a help message

Table 80 Command line parameters for UDP packet generator

The frame types are listed in Table 81.

Frame type	Length	Type
0	70	Beacon message
1	52	Short AX.25 encapsulated frame
2	572	Long AX.25 encapsulated frame
3	900	Huge AX.25 encapsulated frame
4	1200	Large IP encapsulated packet
5	1410	Huge IP encapsulated packet

Table 81 Generated frame types